

# LAB 3 : TokTickIT Authenticated IT Ticketing System

Final submission report — authenticated users, IT Staff ticketing, Administrator safety, and release evidence

|                                    |                                               |
|------------------------------------|-----------------------------------------------|
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## Answer Part 1: Git Use with Engineering Workflow (10 Points)

Lab 3 was developed on issue-specific branches, reviewed through pull requests, integrated into lab3-staging, and released to main. The final evidence/report change was approved and merged as PR #49, merge commit b45a853.

| Evidence                      | Link / result                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------|
| GitHub repository             | HolyThiccDaddy/toktickit                                                                   |
| GitHub project board          | TokTickIT Individual Sprints                                                               |
| Issues #36–#39                | Authentication/migration, requester regression, staff queue/detail, Administrator evidence |
| Implementation release        | PR #46 merged into main as 92fac36                                                         |
| Documentation/evidence review | PR #47 and PR #48 merged before the final report refresh                                   |
| Final report/PDF merge        | PR #49 merged into main as b45a853                                                         |

The canonical report and PDF are the un-suffixed LAB\_03\_FINAL\_SUBMISSION\_REPORT files. The merged main branch is the source of truth for grading; the final evidence commit is 29dd2f8, merged by PR #49.

## Answer Part 1: Git Use with Engineering Workflow (continued)

### Peer-review and release history

| Stage                        | Pull request and result                                        |
|------------------------------|----------------------------------------------------------------|
| Contract                     | PR #40 approved and merged as 40ca398.                         |
| Authentication and migration | PR #41 merged as b49c8b2 after credential and path-gate fixes. |
| Requester regression         | PR #42 merged as 0ee0002.                                      |
| Staff queue                  | PR #43 merged as c1cd708.                                      |
| Administrator implementation | PR #44 approved after requested changes and merged as bd5872a. |
| Documentation follow-up      | PR #47 and PR #48 merged; seed-safety clarification retained.  |
| Final report/PDF             | PR #49 approved and merged into main as b45a853.               |

The reviewer record identifies requested changes, responses, approvals, and merge commits. The current canonical report is the result of the merged PR #49, not a stale working-tree-only artifact.

## Answer Part 2: Spec DD (5 Points)

The approved contract is maintained in the four linked source records below. PR #40 records the contract approval before implementation.

| Document         | Purpose and evidence                                                                                   |
|------------------|--------------------------------------------------------------------------------------------------------|
| specification.md | Requirements, roles, business rules, data model, migration, acceptance criteria and Definition of Done |
| api-spec.md      | Named request/response schemas, status codes, authorization and compatibility                          |
| ui-spec.md       | Screens, states, Zen Green tokens, accessibility and responsive rules                                  |
| tests.md         | AC-to-test traceability and release checks                                                             |

The master specification crosswalk names FR-01–FR-16, BR-01–BR-14, AC-01–AC-15, the authorization matrix, migration decisions, API/UI contracts and the Definition of Done. The complete sources remain linked in the Markdown report.

### Key contract decisions

| Decision               | Accepted rule                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Authenticated boundary | The session replaces the Lab 2 Development Requester selector; invalid, inactive and first-login states are explicit. |
| Authorization          | Requester ownership, IT Staff queue/detail access and Administrator-only user management are server-enforced.         |
| Migration              | Lab 2 identifiers and relationships are preserved; credential backfill and seed behavior are deterministic.           |
| UI                     | Zen Green tokens, role-aware navigation, accessible feedback and no page-level horizontal overflow are required.      |

## Answer Part 3: Test DD and Traceability (10 Points)

The final-main verification was rerun from merged main @ 69ed72e before the documentation/report merge PR #49. PR #49 is documentation/PDF-only and does not change the released application behavior.

| Check                    | Observed result                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Server regression        | 16 test files / 92 tests passed; 0 failed and 0 skipped                                                                  |
| Client regression        | 15 test files / 50 tests passed; 0 failed and 0 skipped                                                                  |
| Server and client builds | Both npm run build commands passed                                                                                       |
| Playwright E2E           | 13 passed, 0 failed, 0 skipped, 0 flaky across desktop/tablet/mobile                                                     |
| Migration                | 5 migrations found; no pending migrations to apply                                                                       |
| Seed                     | Two runs passed after explicit PostgreSQL _test database preflight; each seeded 4 categories, 7 systems and 5 requesters |
| Repository hygiene       | git diff --check passed; local credentials remain ignored                                                                |

Seed safety: npm run prisma:seed has no built-in database guard. Before the recorded runs, DATABASE\_URL was loaded from server/.env.test, checked for PostgreSQL and a database name ending in \_test, set explicitly, and confirmed by Prisma as toktickit\_test.

The full AC-to-test matrix is in tests.md. The report names executable API, unit, UI, security, migration, responsive and E2E paths rather than treating screenshots as test substitutes.

## Answer Part 4: AI Use with Reflection (5 Points)

The LLM used was OpenAI Codex (GPT-5). The selected prompts covered contract extraction, implementation, regression tests, responsive UI, peer-review fixes and final release auditing.

| Prompt focus                | Accepted outcome                                                                                            |
|-----------------------------|-------------------------------------------------------------------------------------------------------------|
| Lab scope and evidence      | Converted the handout into numbered requirements and kept excluded work out of the implementation.          |
| Authentication and security | Accepted session boundaries, CSRF, expiry, logout, inactive-account handling and first-login password gate. |
| Authorization matrix        | Accepted server-side role/ownership checks; rejected client-only or forged-identity shortcuts.              |
| Migration and seed          | Preserved IDs/relationships, deterministic backfill and explicit test-database preflight.                   |
| Queue and detail            | Added searchable/filterable queue, assignment, status/priority and public/internal communication evidence.  |
| Responsive/accessibility    | Mapped tokens, breakpoints, labels, focus and no-overflow checks to executable paths.                       |
| Final audit                 | Corrected stale counts and provenance, then reran final-main tests/builds/E2E/migration/seed.               |

Human decisions and peer review remained authoritative. Suggestions that weakened server-side checks, exposed credentials, relied on hidden UI state or added excluded features were changed or rejected. The final evidence was rerun from merged main before PR #49.

## Answer Part 5: Working Login and Password Change UI (5 Points)

The authenticated session boundary replaces the Lab 2 Development Requester selector. These real local Playwright states show valid and invalid login, inactive-account handling, the first-login password gate, password change, role-aware navigation, logout and direct-access protection.

| 01. Invalid login                                                                                                                                                                                                                                                                                                                                                                                  | 02. Inactive account                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><p><b>TokTickIT</b></p><h3>Sign in to IT Service Desk</h3><p>Use your TokTickIT account to view and manage your requests.</p><div>Invalid email or password</div><p>Email</p><input type="text" value="not-a-user@example.com"/><p>Password</p><input type="password" value="....."/><div>Show</div><p><a href="#">Retry sign in</a></p><div>Sign in</div></div> <p>Safe error and retry.</p> | <div><p><b>TokTickIT</b></p><h3>Sign in to IT Service Desk</h3><p>Use your TokTickIT account to view and manage your requests.</p><div>This account is inactive</div><p>Email</p><input type="text" value="inactive.staff@example.com"/><p>Password</p><input type="password" value="....."/><div>Show</div><p><a href="#">Retry sign in</a></p><div>Sign in</div></div> <p>Seeded inactive account is refused.</p>                                        |
| 03. Valid credentials                                                                                                                                                                                                                                                                                                                                                                              | 04. First-login gate                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <div><p><b>TokTickIT</b></p><h3>Sign in to IT Service Desk</h3><p>Use your TokTickIT account to view and manage your requests.</p><p>Email</p><input type="text" value="admin@example.com"/><p>Password</p><input type="password" value="....."/><div>Show</div><div>Sign in</div></div> <p>Active Administrator entered; password masked.</p>                                                     | <div><p><b>TokTickIT</b></p><h3>Change your password</h3><p>Your account requires a password change before you can access tickets.</p><p>Current password</p><input type="password"/><p>New password</p><input type="password"/><p>Use 12-128 characters.</p><p>Confirm new password</p><input type="password"/><div>Update password Sign out</div><p>Signed in as admin@example.com</p></div> <p>Password change is required before workspace access.</p> |

## Answer Part 5: Working Login and Password Change UI (continued)

### 05. Password-change form

**TokTickIT**

#### Change your password

Your account requires a password change before you can access tickets.

Current password

New password

Use 12-128 characters.

Confirm new password

Update passwordSign out

Signed in as admin@example.com

Current, new and confirmation values remain masked.

### 07. After logout and direct visit

**TokTickIT**

#### Sign in to IT Service Desk

Use your TokTickIT account to view and manage your requests.

Email

Password

Show

Sign in

Protected workspace is unavailable; Sign in returns.

### 06. Password saved — authenticated Administrator workspace

**TokTickIT** User Management Ticket Queue TokTickIT Administrator ADMIN Sign out

## User Management

Manage roles and account activation without exposing credentials.

Search users

Role

Name or email

All roles

Create user

| Display name | Email                   | Role     | State                              | Actions                                              |
|--------------|-------------------------|----------|------------------------------------|------------------------------------------------------|
| Alex Staff   | alex.staff@example.com  | IT_STAFF | Active<br>Password change required | <div>Edit Deactivate</div> <div>Reset password</div> |
| Casey Staff  | casey.staff@example.com | IT_STAFF | Active<br>Password change required | <div>Edit Deactivate</div> <div>Reset password</div> |

Role-aware navigation, User Management, ADMIN badge and Sign out are visible after the password change.

## Answer Part 6: Working IT Staff Ticket Queue UI (5 Points)

IT Staff and Administrators share a searchable queue with status, IT Priority, assignee, category, sorting, pagination, realistic empty/no-results states, retryable failure feedback and responsive cards.

| Queue criterion                             | Direct evidence                                |
|---------------------------------------------|------------------------------------------------|
| Loaded queue and result count               | Fig 6.1; staff-ticket-flow E2E                 |
| Search, status/priority filters and sorting | Fig 6.2; queue API/component tests             |
| No results and clear filters                | Fig 6.3; component tests                       |
| API error and retry                         | Fig 6.4; explicitly simulated 503 interception |
| Page 1/2 and assigned/unassigned            | Figs 6.5–6.7; staff-ticket-flow E2E            |
| Mobile cards and sorting                    | staff-ticket-flow E2E; responsive E2E          |

Figure 6.1. Loaded shared queue

TokTickIT
Ticket Queue
Alex Staff  
IT\_STAFF
Sign out

### Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue
Evidence queue ticket
Status
All statuses
IT Priority
All priorities
Assignee ID
Any assignee
Category
All categories

Sort by
Updated date
Sort direction
Descending
Results per page
20
Clear

| Ticket No.      | Updated ↓             | Status | Summary                  | Requester     | IT Priority | Assignee   | Actions |
|-----------------|-----------------------|--------|--------------------------|---------------|-------------|------------|---------|
| TKT-2026-000027 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 21 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000026 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 20 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000025 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 19 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000024 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 18 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000023 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 17 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000022 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 16 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000021 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 15 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000020 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 14 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000019 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 13 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000018 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 12 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000017 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 11 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000016 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 10 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000015 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 09 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000014 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 08 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000013 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 07 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000012 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 06 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000011 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 05 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000010 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 04 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000009 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 03 | Sarah Johnson | MEDIUM      | Unassigned | Open    |
| TKT-2026-000008 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 02 | Sarah Johnson | MEDIUM      | Unassigned | Open    |

21 results · Page 1 of 2

Previous
1
2
Next

Real local UI capture from artifacts/lab-03/screenshots/staff-flow/states/01\_queue\_loaded.png. Figure 6.4 is explicitly labeled as a Playwright-intercepted 503 simulation.

Figure 6.2. Filtered and sorted queue

TokTickIT

Ticket Queue

Alex Staff  
IT\_STAFF  
Sign out

Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue

Ticket, summary, requester, or ca

Status

NEW

IT Priority

HIGH

Assignee ID

Any assignee

Category

All categories

Sort by

Ticket Number

Sort direction

Descending

Results per page

20

Clear

| Ticket No. ↓    | Updated               | Status | Summary                          | Requester     | IT Priority | Assignee   | Actions |
|-----------------|-----------------------|--------|----------------------------------|---------------|-------------|------------|---------|
| TKT-2026-000006 | 9/29/2026, 8:48:28 PM | NEW    | Staff queue E2E regression       | Sarah Johnson | HIGH        | Unassigned | Open    |
| TKT-2026-000005 | 9/29/2026, 8:48:26 PM | NEW    | Issue 39 visual desktop          | Sarah Johnson | HIGH        | Unassigned | Open    |
| TKT-2026-000002 | 9/29/2026, 8:48:14 PM | NEW    | E2E authenticated requester flow | David Lee     | HIGH        | Unassigned | Open    |

3 results · Page 1 of 1

Previous1Next

Real local UI capture from artifacts/lab-03/screenshots/staff-flow/states/02\_queue\_filter\_sort.png. Figure 6.4 is explicitly labeled as a Playwright-intercepted 503 simulation.

Figure 6.3. No matching results

TokTickIT

Ticket Queue

Alex Staff  
IT\_STAFF  
Sign out

Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue

Status

IT Priority

Assignee ID

Category

no-such-ticket-evidence-2026

NEW

HIGH

Any assignee

All categories

Sort by

Sort direction

Results per page

Clear

Ticket Number

Descending

20

No matching tickets found.

Clear filters

Real local UI capture from artifacts/lab-03/screenshots/staff-flow/states/03\_queue\_no\_results.png. Figure 6.4 is explicitly labeled as a Playwright-intercepted 503 simulation.

Figure 6.4. Retryable API failure

TokTickIT

Ticket Queue

Alex Staff  
IT\_STAFFSign out

Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue

temporary outage evidence

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

Sort by

Ticket Number

Sort direction

Descending

Results per page

20

Clear

Unable to load ticket queue

Queue temporarily unavailable

Retry

Real local UI capture from artifacts/lab-03/screenshots/staff-flow/states/04\_queue\_retry\_simulated\_api\_failure.png. Figure 6.4 is explicitly labeled as a Playwright-intercepted 503 simulation.

## Figure 6.5. Pagination: page 1 of 2

21 results · Page 1 of 2

Previous

1

2

Next

Real local UI capture from artifacts/lab-03/screenshots/staff-flow/states/05\_queue\_page\_1\_of\_2.png. Figure 6.4 is explicitly labeled as a Playwright-intercepted 503 simulation.

Figure 6.6. Pagination: page 2 of 2

TokTickIT

Ticket Queue

Alex Staff  
IT\_STAFF  
Sign out

### Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue

Status

IT Priority

Assignee ID

Category

Evidence queue ticket

All statuses

All priorities

Any assignee

All categories

Sort by

Sort direction

Results per page

Clear

Updated date

Descending

20

| Ticket No.      | Updated ↓             | Status | Summary                  | Requester     | IT Priority | Assignee   | Actions |
|-----------------|-----------------------|--------|--------------------------|---------------|-------------|------------|---------|
| TKT-2026-000007 | 9/29/2026, 8:48:28 PM | NEW    | Evidence queue ticket 01 | Sarah Johnson | MEDIUM      | Unassigned | Open    |

21 results · Page 2 of 2

Previous

1

2

Next

Real local UI capture from artifacts/lab-03/screenshots/staff-flow/states/06\_queue\_page\_2\_of\_2.png. Figure 6.4 is explicitly labeled as a Playwright-intercepted 503 simulation.

Figure 6.7. Assigned and unassigned rows

TokTickIT

Ticket Queue

Alex Staff  
IT\_STAFFSign out

Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue

Ticket, summary, requester, or ca

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

Sort by

Ticket Number

Sort direction

Ascending

Results per page

20

Clear

| Ticket No. I    | Updated               | Status   | Summary                            | Requester         | IT Priority | Assignee    | Actions |
|-----------------|-----------------------|----------|------------------------------------|-------------------|-------------|-------------|---------|
| TKT-2026-000001 | 9/29/2026, 8:48:12 PM | NEW      | Authenticated ownership isolation  | Jennifer Anderson | LOW         | Unassigned  | Open    |
| TKT-2026-000002 | 9/29/2026, 8:48:14 PM | NEW      | E2E authenticated requester flow   | David Lee         | HIGH        | Unassigned  | Open    |
| TKT-2026-000003 | 9/29/2026, 8:48:15 PM | NEW      | Responsive desktop ticket          | Jennifer Anderson | MEDIUM      | Unassigned  | Open    |
| TKT-2026-000004 | 9/29/2026, 8:48:20 PM | NEW      | Requester communication regression | Sarah Johnson     | MEDIUM      | Unassigned  | Open    |
| TKT-2026-000005 | 9/29/2026, 8:48:26 PM | NEW      | Issue 39 visual desktop            | Sarah Johnson     | HIGH        | Unassigned  | Open    |
| TKT-2026-000006 | 9/29/2026, 8:48:32 PM | RESOLVED | Staff queue E2E regression         | Sarah Johnson     | URGENT      | Casey Staff | Open    |

Real local UI capture from artifacts/lab-03/screenshots/staff-flow/states/07\_queue\_assigned\_unassigned.png. Figure 6.4 is explicitly labeled as a Playwright-intercepted 503 simulation.

## Answer Part 7: Working IT Staff Ticket Detail UI (10 Points)

Staff Ticket Detail supports claim/reassign, IT Priority, permitted status transitions and confirmations, attachment continuity, Public Comments, Internal Notes, Requester resolution indication, role restrictions and safe failure behavior.

| Detail criterion                          | Direct evidence                          |
|-------------------------------------------|------------------------------------------|
| Attachment continuity and ticket identity | Figs 7.1; requester-to-staff E2E         |
| Claim/reassign owner                      | Figs 7.2 and 7.7; queue/detail API tests |
| Priority and legal status transitions     | Figs 7.3–7.4; BR-08 API matrix           |
| Confirmation before RESOLVED              | Fig 7.3; blocked then confirmed E2E      |
| Public comment and internal note          | Figs 7.5–7.6; visibility tests           |
| Requester denied Staff access             | Fig 7.8; actual HTTP 403 response        |

Figure 7.1. Attachment continuity

### Attachments

**01\_ticket\_detail\_readonly.png**

245 KB · Uploaded 9/29/2026, 8:48:28 PM

Download

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/10\_attachment\_continuity.png.

Figure 7.2. Claimed owner and operations

### Ticket operations

Assignee ID

6

Claim for me

Save assignment

IT Priority

HIGH

▼

Save

Move status

Select an allowed transition

▼

☐ Confirm transition

Update status

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/11\_claimed\_operation.png.

Figure 7.3. Explicit transition confirmation

TokTickIT

Ticket QueueTicket Detail

Alex Staff  
IT\_STAFFSign out

[← Back to queue](#)

IN\_PROGRESS

Confirm this status transition before saving.

Ticket No.  
**TKT-2026-000006**

URGENT

REQUESTER  
Sarah Johnson (sarah.johnson@example.com)

CATEGORY  
Network

RELATED SYSTEM  
VPN

CREATED  
9/29/2026, 8:48:28 PM

UPDATED  
9/29/2026, 8:48:32 PM

**Staff queue E2E regression**  
Verify that IT Staff can find and open a shared ticket.

**Ticket operations**

Assignee ID  

Claim for meSave assignment

IT Priority  
URGENT

▼

Save

Move status  
RESOLVED (confirmation required)

▼

☐ Confirm transition

Update status

**Attachments**

01\_ticket\_detail\_readonly.png  
245 KB · Uploaded 9/29/2026, 8:48:28 PM

Download

**Public comments**

No public comments yet.

Add a public comment

Post public comment

**Internal notes** Staff only

No internal notes yet.

Add an internal note

Add internal note

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/07\_detail\_confirmation\_required.png.

Figure 7.4. Resolved and urgent

TokTickIT

Ticket QueueTicket Detail

Alex Staff  
IT\_STAFFSign out

[← Back to queue](#)

RESOLVED

Ticket No.  
**TKT-2026-000006**

URGENT

REQUESTER  
Sarah Johnson (sarah.johnson@example.com)

CATEGORY  
Network

RELATED SYSTEM  
VPN

CREATED  
9/29/2026, 8:48:28 PM

UPDATED  
9/29/2026, 8:48:32 PM

**Staff queue E2E regression**  
Verify that IT Staff can find and open a shared ticket.

**Ticket operations**

Assignee ID  
7

Claim for me

Save assignment

IT Priority  
URGENT

Save

Move status  
Select an allowed transition

☐ Confirm transition

Update status

**Attachments**

01\_ticket\_detail\_readonly.png  
245 KB · Uploaded 9/29/2026, 8:48:28 PM

Download

**Public comments**

No public comments yet.

Add a public comment

Post public comment

**Internal notes** Staff only

No internal notes yet.

Add an internal note

Add internal note

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/08\_detail\_resolved.png.

Figure 7.5. Public comment

Public comments

Alex Staff · 9/29/2026, 8:48:32 PM

Public E2E evidence: requester can read this update.

Add a public comment

Post public comment

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/14\_public\_comment.png.

Figure 7.6. Internal note

Internal notes

Staff only

Alex Staff · 9/29/2026, 8:48:33 PM

Internal E2E evidence: staff-only diagnostic note.

Add an internal note

Add internal note

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/15\_internal\_note.png.

Figure 7.7. Reassigned to another IT Staff user

Ticket operations

Assignee ID

7

Claim for me

Save assignment

IT Priority

HIGH

Save

Move status

Select an allowed transition

☐ Confirm transition

Update status

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/16\_reassigned\_operation.png.

### Figure 7.8. Requester denied Staff queue access

Pretty-print ☐

```
{"error":{"code":"FORBIDDEN","message":"Only IT Staff or Administrators may access this resource"}}
```

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/19\_requester\_staff\_403\_response.png.

## Answer Part 8: Working Administrator User Management UI (5 Points)

The Administrator-only screen lists users, supports name/email search and role filtering, creates and edits one-role accounts, changes activation state, resets initial passwords and enforces duplicate-email, self and last-Administrator protections.

| Administrator criterion                  | Direct evidence                       |
|------------------------------------------|---------------------------------------|
| List, search and role filter             | Figs 8.1–8.3; admin UI/API tests      |
| Create and initial password              | Figs 8.4–8.5; user-administration E2E |
| Edit and deactivate                      | Figs 8.6–8.7; user-administration E2E |
| Reset and next-login password gate       | Figs 8.8 and 8.11; admin E2E          |
| Self/last-admin and duplicate safeguards | Figs 8.9–8.10; users-admin API tests  |
| Invalid input and non-admin denial       | Figs 8.12–8.13; UI/API tests          |

Figure 8.1. User list with active states

TokTickIT

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

Name or email

Role

All roles

Create user

| Display name            | Email                         | Role      | State                                | Actions                                 |
|-------------------------|-------------------------------|-----------|--------------------------------------|-----------------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active                               | <div>EditDeactivateReset password</div> |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active                               | <div>EditDeactivateReset password</div> |
| David Lee               | david.lee@example.com         | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | <div>EditDeactivateReset password</div> |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/01\_user\_list.png.

Figure 8.2. Search by email

TokTickIT

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

alex.staff@example.com

Role

All roles

Create user

| Display name | Email                  | Role     | State  | Actions                                 |
|--------------|------------------------|----------|--------|-----------------------------------------|
| Alex Staff   | alex.staff@example.com | IT_STAFF | Active | <div>EditDeactivateReset password</div> |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/02\_user\_search.png.

Figure 8.3. Role filter

**TokTickIT**

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

Role

IT\_STAFF

Create user

| Display name   | Email                      | Role     | State                                | Actions                                 |
|----------------|----------------------------|----------|--------------------------------------|-----------------------------------------|
| Alex Staff     | alex.staff@example.com     | IT_STAFF | Active                               | <div>EditDeactivateReset password</div> |
| Casey Staff    | casey.staff@example.com    | IT_STAFF | Active                               | <div>EditDeactivateReset password</div> |
| Inactive Staff | inactive.staff@example.com | IT_STAFF | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Jamie Staff    | jamie.staff@example.com    | IT_STAFF | Active<br>Password change required   | <div>EditDeactivateReset password</div> |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/03\_user\_role\_filter.png.

Figure 8.4. Create user form

TokTickIT
User Management
Ticket Queue
TokTickIT Administrator
ADMIN
Sign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

Name or email

Role

All roles

Create user

Create user

Cancel

Email

e2e-admin-created@example.com

Display name

E2E Admin Created

Role

IT\_STAFF

☒ Active account

Initial password

.....

12-128 characters; the user must change it on first login.

Create user

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/05\_create\_filled.png.

Figure 8.5. Create success

TokTickIT

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

Created E2E Admin Created.

Search users

Name or email

Role

All roles

Create user

| Display name            | Email                         | Role      | State                                | Actions                                 |
|-------------------------|-------------------------------|-----------|--------------------------------------|-----------------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active                               | <div>EditDeactivateReset password</div> |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active                               | <div>EditDeactivateReset password</div> |
| David Lee               | david.lee@example.com         | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| E2E Admin Created       | e2e-admin-created@example.com | IT_STAFF  | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | <div>EditDeactivateReset password</div> |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/06\_created\_success.png.

Figure 8.6. Edit account

TokTickIT
User Management
Ticket Queue
TokTickIT Administrator
ADMIN
Sign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

Name or email

Role

All roles

Create user

Edit user

Cancel

Email

e2e-admin-updated@example.com

Display name

E2E Admin Updated

Role

REQUESTER

☒ Active account

Save changes

| Display name            | Email                         | Role      | State                                | Actions                                                         |
|-------------------------|-------------------------------|-----------|--------------------------------------|-----------------------------------------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active                               | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active                               | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| David Lee               | david.lee@example.com         | REQUESTER | Active                               | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| E2E Admin Created       | e2e-admin-created@example.com | IT_STAFF  | Active<br>Password change required   | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | <div>Edit</div> <div>Activate</div> <div>Reset password</div>   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active                               | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active                               | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | <div>Edit</div> <div>Activate</div> <div>Reset password</div>   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active                               | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | <div>Edit</div> <div>Deactivate</div> <div>Reset password</div> |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/07\_edit\_form.png.

Figure 8.7. Deactivated account

TokTickIT

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

E2E Admin Updated is now inactive.

Search users

Role

Name or email

All roles

Create user

| Display name            | Email                         | Role      | State                                | Actions                                 |
|-------------------------|-------------------------------|-----------|--------------------------------------|-----------------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active                               | <div>EditDeactivateReset password</div> |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active                               | <div>EditDeactivateReset password</div> |
| David Lee               | david.lee@example.com         | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| E2E Admin Updated       | e2e-admin-updated@example.com | REQUESTER | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active                               | <div>EditDeactivateReset password</div> |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | <div>EditDeactivateReset password</div> |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/09\_deactivated.png.

Figure 8.8. Initial-password reset

TokTickIT
User Management
Ticket Queue
TokTickIT Administrator
ADMIN
Sign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

Reset the initial password for E2E Admin Updated.

Search users
Name or email

Role
All roles

Create user

| Display name            | Email                         | Role      | State                                | Actions                        |
|-------------------------|-------------------------------|-----------|--------------------------------------|--------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active                               | Edit Deactivate Reset password |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active                               | Edit Deactivate Reset password |
| David Lee               | david.lee@example.com         | REQUESTER | Active                               | Edit Deactivate Reset password |
| E2E Admin Updated       | e2e-admin-updated@example.com | REQUESTER | Inactive<br>Password change required | Edit Activate Reset password   |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | Edit Activate Reset password   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | Edit Deactivate Reset password |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active                               | Edit Deactivate Reset password |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active                               | Edit Deactivate Reset password |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | Edit Activate Reset password   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active                               | Edit Deactivate Reset password |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | Edit Deactivate Reset password |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/11\_reset\_success.png.

Figure 8.9. Self-deactivation rejected

TokTickIT
User Management
Ticket Queue
TokTickIT Administrator
ADMIN
Sign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

An Administrator cannot deactivate or change their own role

Retry

Search users
Name or email

Role
All roles

Create user

| Display name            | Email                         | Role      | State                                | Actions                        |
|-------------------------|-------------------------------|-----------|--------------------------------------|--------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active                               | Edit Deactivate Reset password |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active                               | Edit Deactivate Reset password |
| David Lee               | david.lee@example.com         | REQUESTER | Active                               | Edit Deactivate Reset password |
| E2E Admin Updated       | e2e-admin-updated@example.com | REQUESTER | Inactive<br>Password change required | Edit Activate Reset password   |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | Edit Activate Reset password   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | Edit Deactivate Reset password |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active                               | Edit Deactivate Reset password |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active                               | Edit Deactivate Reset password |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | Edit Activate Reset password   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active                               | Edit Deactivate Reset password |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | Edit Deactivate Reset password |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/12\_self\_deactivation\_rejected.png.

Figure 8.10. Duplicate email rejected

TokTickIT

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

A user with this email already exists

Retry

Search users

Name or email

Role

All roles

Create user

Create user

Cancel

Email

e2e-admin-updated@example.com

Display name

Duplicate E2E User

Role

REQUESTER

☒ Active account

Initial password

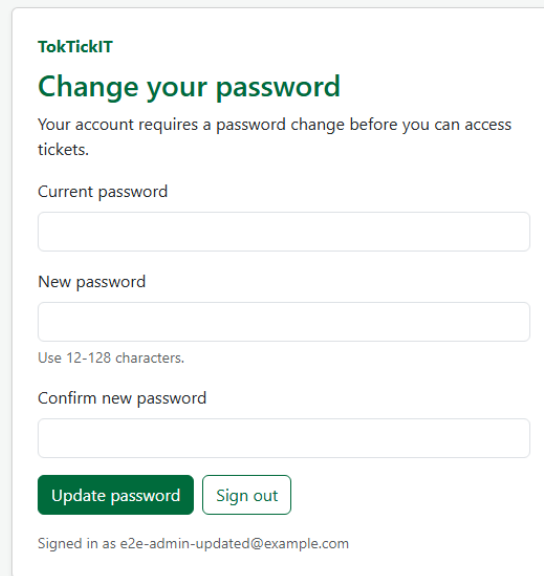
.....

12-128 characters; the user must change it on first login.

Create user

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/13\_duplicate\_email\_rejected.png.

**Figure 8.11. Reset requires password change on next login**



The screenshot shows a web form titled 'TokTickIT Change your password'. The form is set against a light gray background. It contains three text input fields: 'Current password', 'New password', and 'Confirm new password'. Below the 'New password' field is a small text hint: 'Use 12-128 characters.' At the bottom of the form are two buttons: a dark green 'Update password' button and a light green 'Sign out' button. Below the buttons, it says 'Signed in as e2e-admin-updated@example.com'.

**TokTickIT**

## Change your password

Your account requires a password change before you can access tickets.

Current password

New password

Use 12-128 characters.

Confirm new password

[Update password](#) [Sign out](#)

Signed in as e2e-admin-updated@example.com

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/14\_reset\_requires\_change\_on\_next\_login.png.

Figure 8.12. Invalid create input blocked

TokTickIT
User Management
Ticket Queue
TokTickIT Administrator
ADMIN
Sign out

## User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users
Role
Name or email
All roles
Create user

### Create user

[Cancel](#)

Email
not-an-email
Display name
Invalid Evidence User

Role
Please include an '@' in the email address. 'not-an-email' is missing an '@'.
REQUESTER
Active account

Initial password
.....
12-128 characters; the user must change it on first login.

Create user

| Display name            | Email                         | Role      | State                                | Actions                        |
|-------------------------|-------------------------------|-----------|--------------------------------------|--------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active                               | Edit Deactivate Reset password |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active                               | Edit Deactivate Reset password |
| David Lee               | david.lee@example.com         | REQUESTER | Active                               | Edit Deactivate Reset password |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | Edit Activate Reset password   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | Edit Deactivate Reset password |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active                               | Edit Deactivate Reset password |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active                               | Edit Deactivate Reset password |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | Edit Activate Reset password   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active                               | Edit Deactivate Reset password |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | Edit Deactivate Reset password |

Real local UI/API capture from artifacts/lab-03/screenshots/user-management/states/04\_invalid\_input\_blocked.png.

**Figure 8.13. IT Staff denied Administrator access**

Pretty-print ☐

```
{"error":{"code":"FORBIDDEN","message":"Only  
Administrators may access user management"}}
```

Real local UI/API capture from artifacts/lab-03/screenshots/staff-flow/states/20\_staff\_admin\_403\_response.png.

## Answer Part 9: Zen Green UI and Responsive Evidence (5 Points)

The released screens reuse the Zen Green tokens and shared controls established in Lab 2. The table below records the approved tokens, breakpoints, accessibility rules and no-overflow acceptance before the original desktop/tablet/mobile captures.

| Zen Green token and responsive rule record |                                                                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Token/rule                                 | Approved value or rule                                                                                                              |
| Primary / secondary / pale green           | #006B3C / #0B7A46 / #EAF6EF                                                                                                         |
| Page / surface / primary text              | #F5F7F6 / #FFFFFF / #1B2F24                                                                                                         |
| Secondary text / border / read-only        | #52665A / #D1DDD6 / #EEF3F0                                                                                                         |
| Error / warning / urgent / low             | #C53030 / #D69E2E / #E53E3E / #319795                                                                                               |
| Breakpoints                                | Desktop >=992px; tablet 768–991px; mobile <768px                                                                                    |
| Responsive rule                            | Reflow without clipping/overlap; tables stack or scroll inside a component only when necessary; no page-level horizontal overflow   |
| Accessibility rule                         | Labels and keyboard operation for every control, associated/live-region errors, visible focus after mutations and readable contrast |

| Completed visual and accessibility checklist |                                                                                |
|----------------------------------------------|--------------------------------------------------------------------------------|
| Check                                        | Result and evidence                                                            |
| Zen Green and type                           | Approved colors, consistent typography and shared controls across roles        |
| Role-aware navigation                        | Login shell and role routes appear in Parts 5–8 and authentication E2E         |
| Badges and status                            | Queue/detail priority and status remain distinguishable                        |
| Editable/read-only                           | Staff/Admin actions are separated from requester read-only fields              |
| Validation and errors                        | Field errors, safe denials and retry placement are captured                    |
| Focus and keyboard                           | Semantic labels and focus behavior are covered by client tests and UI contract |
| Clipping and overlap                         | Visual review of all 12 captured viewport states found none                    |
| Horizontal overflow                          | Playwright checks document width at all 12 view/viewport combinations          |

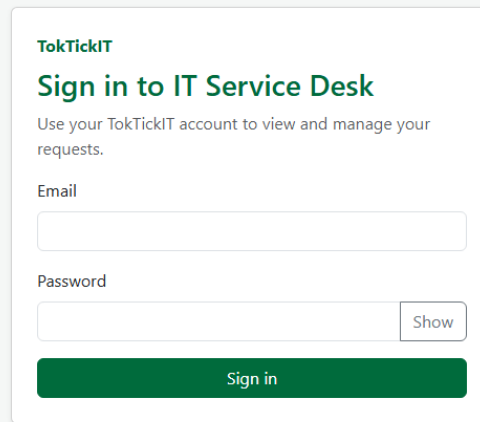
## Answer Part 9: Zen Green UI and Responsive Evidence (continued)

The four Lab 3 views below were captured from the running app in desktop, tablet and mobile Playwright projects against the dedicated test database. All twelve original PNGs are embedded at their source aspect ratio.

| View                | Desktop     | Tablet      | Mobile      |
|---------------------|-------------|-------------|-------------|
| Authentication      | Figure 9.1  | Figure 9.2  | Figure 9.3  |
| Staff Queue         | Figure 9.4  | Figure 9.5  | Figure 9.6  |
| Staff Ticket Detail | Figure 9.7  | Figure 9.8  | Figure 9.9  |
| User Management     | Figure 9.10 | Figure 9.11 | Figure 9.12 |

Playwright E2E: 13 passed, 0 failed, 0 skipped, 0 flaky across desktop, tablet and mobile. The UI spec and client tests cover labels, focus, safe errors, mobile card sorting and no page-level horizontal overflow.

**Figure 9.1. Authentication — desktop**



The screenshot shows a login form for TokTickIT. At the top left is the TokTickIT logo. Below it is the heading "Sign in to IT Service Desk" in a large, bold, dark green font. Under the heading is a line of smaller text: "Use your TokTickIT account to view and manage your requests." Below this text are two input fields. The first is labeled "Email" and is a simple white box with a light gray border. The second is labeled "Password" and is a white box with a light gray border, featuring a "Show" button on its right side. At the bottom of the form is a large, solid dark green button with the text "Sign in" in white.

**TokTickIT**

## Sign in to IT Service Desk

Use your TokTickIT account to view and manage your requests.

Email

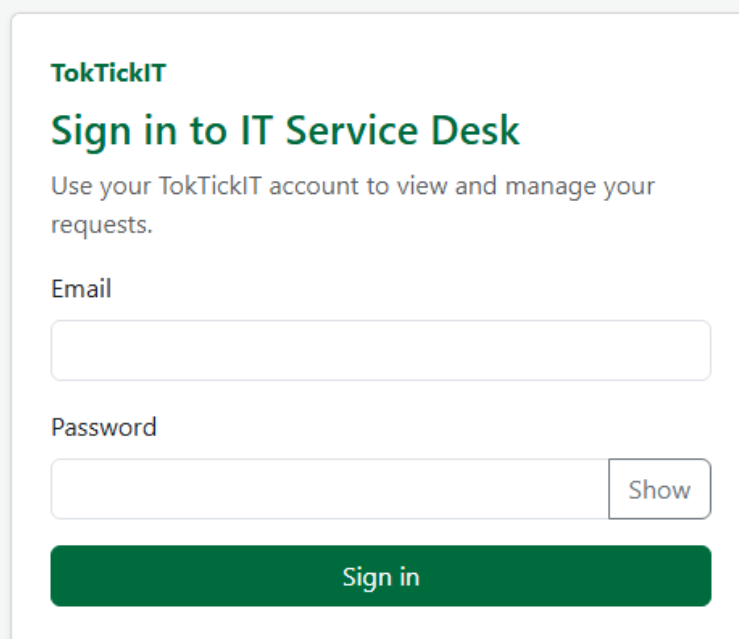
Password

Show

**Sign in**

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/authentication/desktop.png. No UI content was altered.

Figure 9.2. Authentication — tablet



The image shows a tablet screen with a white authentication form centered on a light gray background. The form has a rounded rectangle shape with a subtle drop shadow. At the top left of the form is the 'TokTickIT' logo in a bold, dark green font. Below the logo is the main heading 'Sign in to IT Service Desk' in a large, dark green font. Underneath the heading is a line of smaller, gray text: 'Use your TokTickIT account to view and manage your requests.' The form contains two input fields: an 'Email' field and a 'Password' field, both with light gray borders and rounded corners. To the right of the password field is a small, light gray button labeled 'Show'. At the bottom of the form is a wide, dark green button with the text 'Sign in' in white, centered.

**TokTickIT**

## Sign in to IT Service Desk

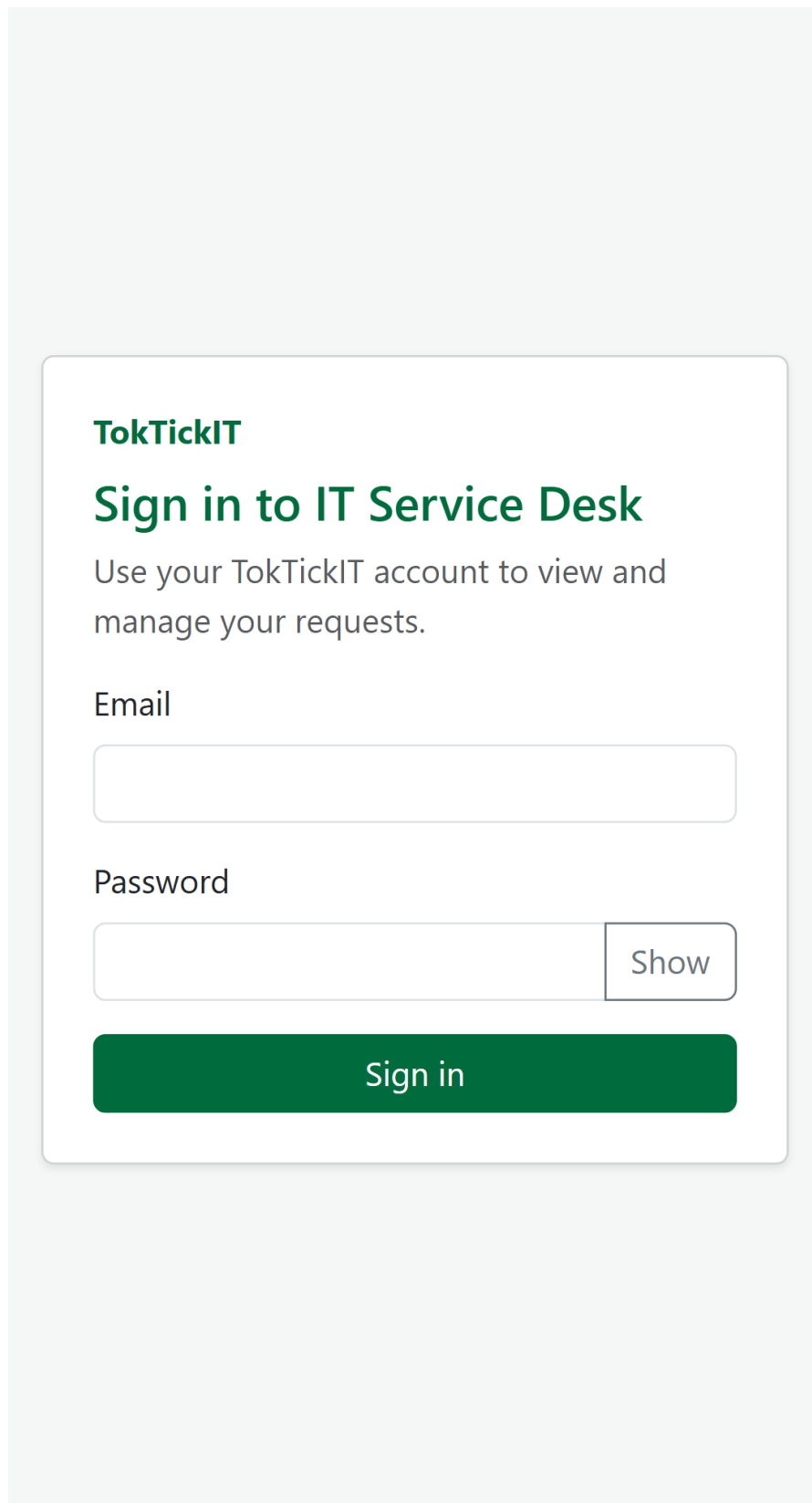
Use your TokTickIT account to view and manage your requests.

Email

Password

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/authentication/tablet.png. No UI content was altered.

Figure 9.3. Authentication — mobile



The image shows a mobile authentication screen for TokTickIT. It features a white card with rounded corners on a light gray background. The card contains the TokTickIT logo, a sign-in title, a descriptive sentence, input fields for email and password, a 'Show' button for password visibility, and a large green 'Sign in' button.

**TokTickIT**

## Sign in to IT Service Desk

Use your TokTickIT account to view and manage your requests.

Email

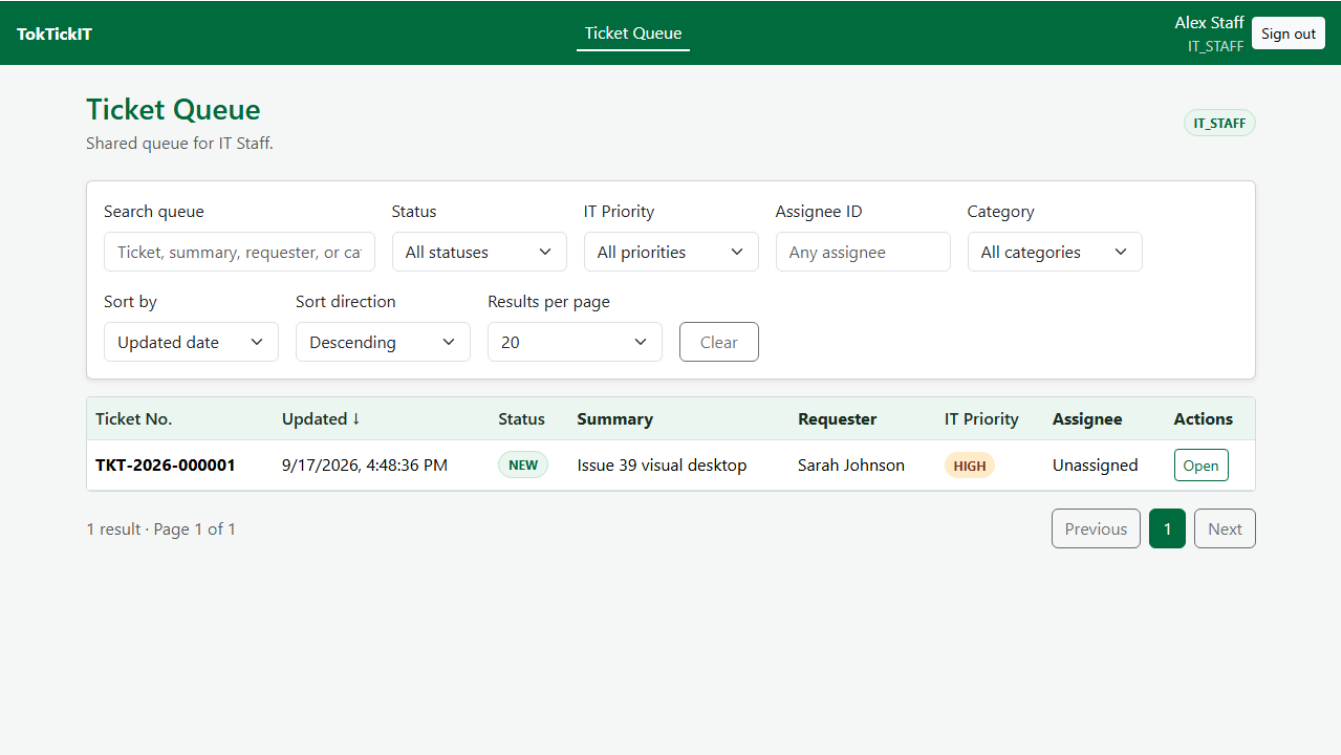
Password

Show

Sign in

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/authentication/mobile.png. No UI content was altered.

Figure 9.4. Staff Queue — desktop



Original full-page Playwright screenshot: artifacts/lab-03/screenshots/staff-queue/desktop.png. No UI content was altered.

Figure 9.5. Staff Queue — tablet

TokTickIT
Ticket Queue
Alex Staff  
IT\_STAFF
Sign out

## Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue

Status
All statuses

IT Priority
All priorities

Assignee ID
Any assignee

Category
All categories

Sort by
Updated date

Sort direction
Descending

Results per page
20

Clear

| Ticket No.             | Updated ↓             | Status | Summary                | Requester     | IT Priority | Assignee   | Actions |
|------------------------|-----------------------|--------|------------------------|---------------|-------------|------------|---------|
| <b>TKT-2026-000001</b> | 9/17/2026, 4:48:58 PM | NEW    | Issue 39 visual tablet | Sarah Johnson | HIGH        | Unassigned | Open    |

1 result · Page 1 of 1

Previous
1
Next

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/staff-queue/tablet.png. No UI content was altered.

Figure 9.6. Staff Queue — mobile

TokTickIT

Ticket Queue

Alex Staff  
IT\_STAFF

Sign out

Ticket Queue

Shared queue for IT Staff.

IT\_STAFF

Search queue

Ticket, summary, requester, or category

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

Sort by

Updated date

Sort direction

Descending

Results per page

20

Clear

TKT-2026-000002

NEW

Issue 39 visual mobile

Requester: Sarah Johnson

HIGH Unassigned

Updated 9/17/2026, 4:49:14 PM

Open ticket

TKT-2026-000001

NEW

Issue 39 visual tablet

Requester: Sarah Johnson

HIGH Unassigned

Updated 9/17/2026, 4:48:58 PM

Open ticket

2 results · Page 1 of 1

Previous

1

Next

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/staff-queue/mobile.png. No UI content was altered.

Figure 9.7. Staff Ticket Detail — desktop

TokTickIT

Ticket QueueTicket Detail

Alex Staff  
IT\_STAFFSign out

[← Back to queue](#)

NEW

Ticket No.  
**TKT-2026-000001**

REQUESTER  
Sarah Johnson (sarah.johnson@example.com)

CATEGORY  
Network

RELATED SYSTEM  
VPN

CREATED  
9/17/2026, 4:48:36 PM

UPDATED  
9/17/2026, 4:48:36 PM

HIGH

**Issue 39 visual desktop**  
Release evidence ticket for the staff detail view.

**Ticket operations**

Assignee ID

Blank to unassign

Claim for me

Save assignment

IT Priority

HIGH

▼

Save

Move status

Select an allowed transition

▼

☐ Confirm transition

Update status

**Attachments**

No attachments on this ticket.

**Public comments**

No public comments yet.

Add a public comment

Post public comment

**Internal notes** **Staff only**

No internal notes yet.

Add an internal note

Add internal note

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/staff-ticket-detail/desktop.png. No UI content was altered.

Figure 9.8. Staff Ticket Detail — tablet

TokTickIT

Ticket QueueTicket Detail

Alex Staff  
IT\_STAFFSign out

[← Back to queue](#)

NEW

Ticket No.  
**TKT-2026-000001**

HIGH

REQUESTER  
Sarah Johnson  
(sarah.johnson@example.com)

CATEGORY  
Network

RELATED SYSTEM  
VPN

CREATED  
9/17/2026, 4:48:58 PM

UPDATED  
9/17/2026, 4:48:58 PM

**Issue 39 visual tablet**  
Release evidence ticket for the staff detail view.

**Ticket operations**

Assignee ID

Blank to unassign

Claim for me

Save assignment

IT Priority

HIGH

Save

Move status

Select an allowed transition

☐ Confirm transition

Update status

**Attachments**

No attachments on this ticket.

**Public comments**

No public comments yet.

Add a public comment

Post public comment

**Internal notes** **Staff only**

No internal notes yet.

Add an internal note

Add internal note

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/staff-ticket-detail/tablet.png. No UI content was altered.

Figure 9.9. Staff Ticket Detail — mobile

TokTickIT

Ticket Queue

Ticket Detail

Alex Staff  
IT\_STAFF

Sign out

[← Back to queue](#)

NEW

Ticket No.

TKT-2026-000002

HIGH

REQUESTER

Sarah Johnson  
(sarah.johnson@example.com)

CATEGORY

Network

RELATED SYSTEM

VPN

CREATED

9/17/2026, 4:49:14 PM

UPDATED

9/17/2026, 4:49:14 PM

Issue 39 visual mobile

Release evidence ticket for the staff detail view.

Ticket operations

Assignee ID

Blank to unassign

Claim for me

Save assignment

IT Priority

HIGH

Save

Move status

Select an allowed transition

☐ Confirm transition

Update status

Attachments

No attachments on this ticket.

Public comments

No public comments yet.

Add a public comment

Post public comment

Internal notes

Staff only

No internal notes yet.

Add an internal note

Add internal note

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/staff-ticket-detail/mobile.png. No UI content was altered.

Figure 9.10. User Management — desktop

TokTickIT

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

Name or email

Role

All roles

Create user

| Display name            | Email                         | Role      | State                                | Actions                                 |
|-------------------------|-------------------------------|-----------|--------------------------------------|-----------------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| David Lee               | david.lee@example.com         | REQUESTER | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | <div>EditActivateReset password</div>   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active<br>Password change required   | <div>EditDeactivateReset password</div> |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | <div>EditDeactivateReset password</div> |

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/user-management/desktop.png. No UI content was altered.

Figure 9.11. User Management — tablet

TokTickIT
User Management
Ticket Queue
TokTickIT Administrator
ADMIN
Sign out

### User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users
Name or email

Role
All roles
Create user

| Display name            | Email                         | Role      | State                                | Actions                              |
|-------------------------|-------------------------------|-----------|--------------------------------------|--------------------------------------|
| Alex Staff              | alex.staff@example.com        | IT_STAFF  | Active<br>Password change required   | Edit<br>Deactivate<br>Reset password |
| Casey Staff             | casey.staff@example.com       | IT_STAFF  | Active<br>Password change required   | Edit<br>Deactivate<br>Reset password |
| David Lee               | david.lee@example.com         | REQUESTER | Active<br>Password change required   | Edit<br>Deactivate<br>Reset password |
| Inactive Staff          | inactive.staff@example.com    | IT_STAFF  | Inactive<br>Password change required | Edit<br>Activate<br>Reset password   |
| Jamie Staff             | jamie.staff@example.com       | IT_STAFF  | Active<br>Password change required   | Edit<br>Deactivate<br>Reset password |
| Jennifer Anderson       | jennifer.anderson@example.com | REQUESTER | Active<br>Password change required   | Edit<br>Deactivate<br>Reset password |
| Michael Brown           | michael.brown@example.com     | REQUESTER | Active<br>Password change required   | Edit<br>Deactivate<br>Reset password |
| Robert Taylor           | robert.taylor@example.com     | REQUESTER | Inactive<br>Password change required | Edit<br>Activate<br>Reset password   |
| Sarah Johnson           | sarah.johnson@example.com     | REQUESTER | Active<br>Password change required   | Edit<br>Deactivate<br>Reset password |
| TokTickIT Administrator | admin@example.com             | ADMIN     | Active                               | Edit<br>Deactivate<br>Reset password |

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/user-management/tablet.png. No UI content was altered.

Figure 9.12. User Management — mobile

TokTickITUser ManagementTicket Queue

TokTickIT AdministratorADMINSign out

### User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

Role

All roles

Create user

Alex Staff

alex.staff@example.com

Active

IT\_STAFF

EditDeactivateReset password

Casey Staff

casey.staff@example.com

Active - Password change required

IT\_STAFF

EditDeactivateReset password

David Lee

david.lee@example.com

Active - Password change required

REQUESTER

EditDeactivateReset password

Inactive Staff

inactive.staff@example.com

Inactive - Password change required

IT\_STAFF

EditActivateReset password

Jamie Staff

jamie.staff@example.com

Active - Password change required

IT\_STAFF

EditDeactivateReset password

Jennifer Anderson

jennifer.anderson@example.com

Active - Password change required

REQUESTER

EditDeactivateReset password

Michael Brown

michael.brown@example.com

Active - Password change required

REQUESTER

EditDeactivateReset password

Robert Taylor

robert.taylor@example.com

Inactive - Password change required

REQUESTER

EditActivateReset password

Sarah Johnson

sarah.johnson@example.com

Active

REQUESTER

EditDeactivateReset password

TokTickIT Administrator

admin@example.com

Active

ADMIN

EditDeactivateReset password

Original full-page Playwright screenshot: artifacts/lab-03/screenshots/user-management/mobile.png. No UI content was altered.

TokTickIT Lab 3 | Page 52

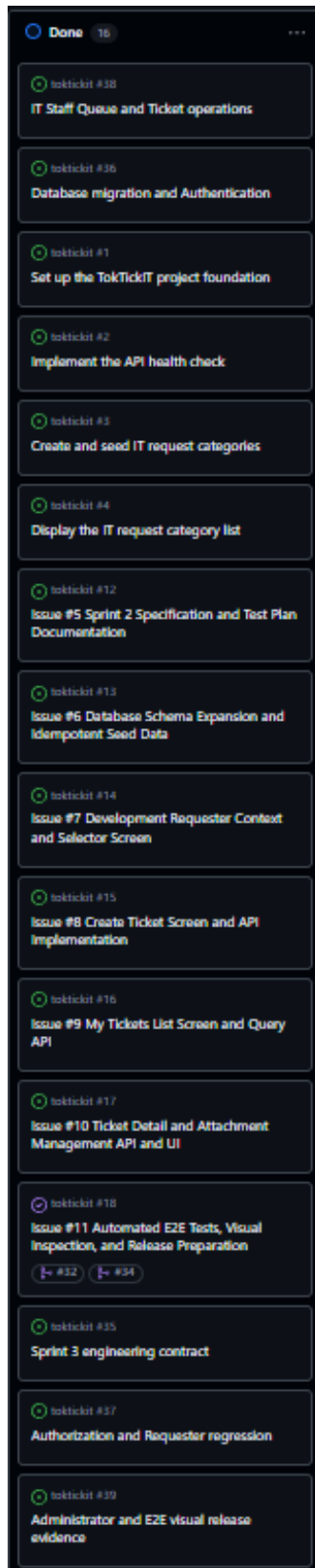
## Evidence Appendix: readable source and verification records

The appendix keeps the source records and terminal evidence readable while preserving the same Lab 2 tables, borders, spacing and numbered footer. The complete Markdown sources remain linked from the report.

| Record            | What the reader can verify                                                                        |
|-------------------|---------------------------------------------------------------------------------------------------|
| specification.md  | Numbered requirements, business/security rules, authorization matrix, migration decisions and DoD |
| api-spec.md       | Request/response schemas, status codes, CSRF/session boundary, authorization and safe errors      |
| ui-spec.md        | Zen Green tokens, breakpoints, accessibility acceptance and no-overflow rules                     |
| tests.md          | AC-to-test mapping, visual checklist and release checks                                           |
| Verification logs | Server/client tests and builds, 13 Playwright tests, migration and two seed runs                  |

The report distinguishes screenshot evidence from executable proof. Screenshots come from the local running app and the dedicated `_test` database; counts are taken from the recorded commands and result files.

## Evidence 01 — Project board with completed Lab 1–3 issues



Original local capture retained under artifacts/lab-03/screenshots/report/01\_kanban\_done.png.

## Evidence 02 — Main-branch commit history

The screenshot displays the GitHub interface for the repository **HolyThiccDaddy / toktickit**. The **Commits** tab is selected, showing the commit history for the **main** branch. The interface includes a search bar at the top right and a navigation bar with links to Code, Issues, Pull requests, Agents, Discussions, Actions, Projects, Wiki, Security and quality, Insights, and Settings.

The commit history is organized by date, with sections for **Commits on Sep 19, 2025**, **Commits on Sep 16, 2025**, **Commits on Sep 17, 2025**, **Commits on Sep 15, 2025**, **Commits on Sep 12, 2025**, and **Commits on Sep 6, 2025**.

Key commits include:

- Merge pull request #47 from HolyThiccDaddy/feature/lab3-final-evidence** (Verified, e79843)
- doc: match Lab 3 report to submission format** (d64133)
- doc: finalize Lab 3 main release evidence** (64b481)
- Merge pull request #46 from HolyThiccDaddy/feature/lab3-final-release** (Verified, 82ac36)
- test: wait for ticket references before submitting** (3b05a2)
- Merge pull request #45 from HolyThiccDaddy/feature/39-admin-e2e-release** (Verified, e82a21)
- doc: record PR 44 review and reflection** (b494b8)
- doc: align Lab 3 evidence records** (baf40)
- doc: update Lab 3 review status** (5c0940)
- Merge pull request #44 from HolyThiccDaddy/feature/39-admin-e2e-release** (Verified, b4c572a)
- test: cover administrator update races** (Verified, 41562d8)
- fix: prevent stale administrator user updates** (Verified, 3891266)
- fix: harden administrator management concurrency and filtering** (b1bc978)
- feat: complete Issue #39 admin and release evidence** (78b4b4)
- Merge pull request #43 from HolyThiccDaddy/feature/38-staff-queue** (Verified, 4e0708)
- fix: address staff queue review feedback** (7994da)
- feat: implement Issue #38 staff queue and ticket operations** (92d07d)
- Merge pull request #42 from HolyThiccDaddy/feature/37-requester-regression** (Verified, 0e0002)
- doc: refresh Lab 3 test totals** (3b3d3b)
- fix: address Lab 3 session and resolution review feedback** (e9cc20)
- feat: migrate requester flows to authenticated sessions** (09c4bc)
- Merge pull request #41 from HolyThiccDaddy/feature/36-db-auth** (Verified, b4b4b2)
- doc: record Lab 3 implementation review** (4cc236f)
- fix: normalize first-login gate paths** (9e0d73a)
- fix: harden Lab 3 seed migration for legacy users** (82c5aa)
- fix: harden Lab 3 auth migration evidence** (2d75d6)
- feat: implement Lab 3 database migration and authentication** (9c053f)
- Merge pull request #40 from HolyThiccDaddy/feature/lab3-contract** (Verified, 40ca398)
- doc: complete Lab 3 contract review updates** (24c7857)
- doc: address Lab 3 contract review feedback** (887c96)
- doc: link Lab 3 contract review PR** (4c98d9)
- doc: add Lab 3 engineering contract** (2a5510a)
- Merge pull request #34 from HolyThiccDaddy/lab2-staging** (Verified, 4b6f53)
- Merge pull request #33 from HolyThiccDaddy/feature/lab2-reviewer-docs** (Verified, 34da9d8)
- doc: reconcile Lab 2 release evidence** (95019b)

At the bottom of the page, there is a footer with copyright information: © 2025 GitHub, Inc. and links to Terms, Privacy, Security, Status, Community, Docs, Contact, and Manage cookies. A note states: "Do not share my personal information".

Original local capture retained under artifacts/lab-03/screenshots/report/02\_main\_commit\_history.png.

## Evidence 03 — PR #40 contract review

Original local capture retained under artifacts/lab-03/screenshots/report/03\_pr40\_contract\_review.png.

## Evidence 04 — PR #46 implementation release

The screenshot shows a GitHub pull request (PR) #46 titled "Release Lab 3 to main". The PR is merged and shows a list of commits and pull requests. The summary section includes a list of verification steps and a release scope. The commit history shows a series of commits related to Lab 3 implementation, including adding engineering contracts, reviewing PRs, and implementing various features like database migration, authentication, and staff queue management. The PR was approved by "otick" and merged into the main branch. The interface includes a sidebar with navigation links, a top bar with search and settings, and a bottom section for adding comments.

**Release Lab 3 to main #46**

Merged 28 commits into [main](#) from [feature/lab3-final-release](#) 1 hour ago

Conversation 3 Commits 28 Checks 0 Files changed 87

**Summary**

- Promote the reviewed Lab 3 implementation and documentation from lab3-staging to main.
- Stabilize the Lab 2 Create Ticket validation test so the release suite is deterministic.

**Verification**

- Server: 15 test files / 62 tests passed twice.
- Client: 15 test files / 50 tests passed twice after the test fix.
- Server and client production builds passed.
- Playwright E2E: 13 tests passed across desktop, tablet, and mobile.
- git diff --check passed.

**Release scope**

- PR [1](#) - [Issue #39](#): Administrator and E2E visual release evidence #44 and PR [1](#) - [Issue #39](#): Update Lab 3 review [otick](#) are already merged into lab3-staging.
- This PR is the final integration of the reviewed Lab 3 work into main.

**Commits**

- [docs: add Lab 3 engineering contract](#) 2d5553a
- [docs: title Lab 3 contract review PR](#) 4d9860c
- [docs: address Lab 3 contract review feedback](#) 887c206
- [docs: complete Lab 3 contract review updates](#) 24c7857
- [large pull request #40 from HolyThiccDaddy/feature/lab3-contract](#) Verified 48c4208
- [test: implement Lab 3 database migration and authentication](#) 5d8025f
- [fix: harden Lab 3 auth migration evidence](#) 2d8756b
- [fix: harden Lab 3 seed migration for legacy users](#) 83e760a
- [fix: normalize first-login gate paths](#) 6ed773a
- [docs: record Lab 3 implementation review](#) 6c7c36f
- [large pull request #41 from HolyThiccDaddy/feature/lab3-auth](#) Verified 645c8b2
- [test: migrate requester flows to authenticated sessions](#) 896c09c
- [fix: address Lab 3 session and registration review feedback](#) e79c088
- [docs: refresh Lab 3 test totals](#) 76b833a
- [large pull request #42 from HolyThiccDaddy/feature/lab3-requester-regis](#) Verified 8ee0802
- [test: implement Issue #38 staff queue and ticket operations](#) 92d871d
- [fix: address staff queue review feedback](#) 76b06da
- [large pull request #43 from HolyThiccDaddy/feature/lab3-staff-queue](#) Verified 13e7766
- [test: complete Issue #39 admin and release evidence](#) 76b0c1c
- [fix: harden administrator management concurrency and filtering](#) 61cc876
- [fix: prevent stale administrator user updates](#) Verified 389326a
- [test: cover administrator update races](#) Verified 415c3ab
- [large pull request #44 from HolyThiccDaddy/feature/lab3-admin-e2e-release](#) Verified 6d8775a
- [docs: update Lab 3 review status](#) 5cd0948
- [docs: align Lab 3 evidence records](#) 13e7f48
- [docs: record PR 44 review and reflection](#) 6ed840b
- [large pull request #45 from HolyThiccDaddy/feature/lab3-admin-e2e-release](#) Verified e52c271
- [test: wait for ticket references before submitting](#) 2d60da2

**Reviews**

otick approved these changes 1 hour ago

**Comments**

otick left a comment

Good bro, wanna merge?

HolyThiccDaddy commented 1 hour ago

Good bro, wanna merge?

Yes, my friend.

otick merged commit 92ac36 into main 1 hour ago

HolyThiccDaddy mentioned this pull request 1 hour ago

docs: finalize Lab 3 main release evidence #47

Pull request successfully merged and closed

You're all set — the [feature/lab3-final-release](#) branch can be safely deleted.

Add a comment

Write Preview

Add your comment here...

Markdown is supported Paste, drop, or click to add files

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#)

ProTip! Add [github](#) or [diff](#) to the end of URLs for Git's plaintext view.

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Original local capture retained under artifacts/lab-03/screenshots/report/04\_pr46\_release\_merge.png.

## Evidence 05 — Peer-review record

```
1 # Lab 3 - Peer Review Record
2
3 Status: Lab 3 contract, implementation, documentation, and release evidence are approved and integrated into 'main'. PR #44 was approved by @osik in [review #5245822348](https://github.com/HolyThiccDaddy/toktickit/pull/44#pullrequestreview-5245822348) and merged into
4
5 **Author:** Thira Rungruangkaset - GitHub: @HolyThiccDaddy
6 **Peer reviewer:** Ashira Sangkaset - GitHub: @osik
7
8 ## Contract review
9
10 | PR | Scope | Reviewer verdict | Follow-up |
11 |---|---|---|---|
12 | #40](https://github.com/HolyThiccDaddy/toktickit/pull/40) | Lab 3 specification, API contract, UI specification, and test plan | Approved by @osik after the second review; merged into 'lab3-staging' as '40ca398' | Contract approval recorded; implementation and rel
13
14 ## Review protocol
15
16 Reviewers should check that the contract covers authentication, first-login password change, role and ownership authorization, staff queue/detail, public comments/internal notes, admin safeguards, migration compatibility, responsive UI, and traceable tests. Record com
17
18 ## Peer review round 1 - 2026-09-12
19
20 The reviewer requested a complete eight-status transition matrix with permitted roles, confirmations, and invalid-transition behavior. The contract now defines NEW, OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CLOSED, REOPENED, and CANCELLED, including terminal
21
22 The reviewer found a contradiction about Administrator Ticket access. The contract now states that the authorization matrix explicitly grants Administrators the same queue and Ticket operations as IT Staff, while keeping user management Administrator-only. The API, UI
23
24 The reviewer requested exact seed quantities, queue query behavior, and Administrator safeguards. The contract now specifies the required active and inactive account counts, realistic assigned and unassigned Tickets, searchable/filterable/sortable fields, pagination m
25
26 The reviewer requested Requester-side tests for comments and the resolution indication. The test plan now maps AC-86 to a Requester Ticket Detail component test and a Requester regression E2E flow that checks ownership rejection and confirms the indication does not set
27
28 The reviewer requested safer inactive-login handling and Administrator assignment eligibility. The API now returns ACCOUNT_INACTIVE only after the submitted password is verified, and assignment accepts only active IT Staff or Administrator users. Public/internal messa
29
30 ## Peer review round 2 - 2026-09-12
31
32 The reviewer requested a global backend first-login gate. The API contract now limits a must-change session to '/auth/me', '/auth/csrf', '/auth/change-password', and '/auth/logout', with 'PASSWORD_CHANGE_REQUIRED' for every other protected endpoint, and the test plan
33
34 The reviewer requested precise API response schemas and success codes. The API contract now defines shared named schemas for sessions, users, tickets, comments, notes, attachments, and pagination, plus a success-status map for every endpoint.
35
36 The reviewer requested complete data-model decisions. The specification now includes a model table with types, nullability, relationships, indexes, deletion behavior, migration mapping, and preservation of Lab 2 identifiers and ownership.
37
38 The reviewer requested the actual IT Priority enum. The contract now defines 'LOW', 'MEDIUM', 'HIGH', and 'URGENT', requires the migrated or newly created Ticket to copy Requested Priority into IT Priority, and maps validation to the API and test plan.
39
40 ## Contract approval - 2026-09-12
41
42 The reviewer approved the corrected contract in [PR #40](https://github.com/HolyThiccDaddy/toktickit/pull/40#pullrequestreview-5186811435). GitHub merged it into 'lab3-staging' as [commit '40ca398'](https://github.com/HolyThiccDaddy/toktickit/commit/40ca3983e786c72d
43
44 ## Implementation review round 1 - 2026-09-12
45
46 In [PR #41](https://github.com/HolyThiccDaddy/toktickit/pull/41#discussion_r3996194389), the reviewer found that seeding only the five hard-coded requester fixtures left additional Lab 2 requesters with the '___MIGRATION_PENDING___' marker. The reviewer also found in [ti
47
48 The reviewer then identified a case-sensitive first-login gate in 'server/src/auth.ts' that could be bypassed with a mixed-case protected path. Commit ['9e9473a'](https://github.com/HolyThiccDaddy/toktickit/commit/9e9473ac499b09ef46e2b2b2f2e07d511ed6c) normalizes r
49
50 ## Implementation and release history
51
52 Issue #38 was merged into 'lab3-staging' through [PR #43](https://github.com/HolyThiccDaddy/toktickit/pull/43). Issue #39 added Administrator user management and the remaining release evidence on 'feature/39-admin-e2e-release'. The implementation includes the Administ
53
54 The final Issue #38 verification is recorded in 'docs/lab-03/tests.md': 15 server test files/92 tests, 15 client test files/50 tests, 13 Playwright tests across desktop/tablet/mobile projects, and production builds passed. The reviewed implementation and documentation
```

Original local capture retained under artifacts/lab-03/screenshots/report/05\_reviewer\_record.png.

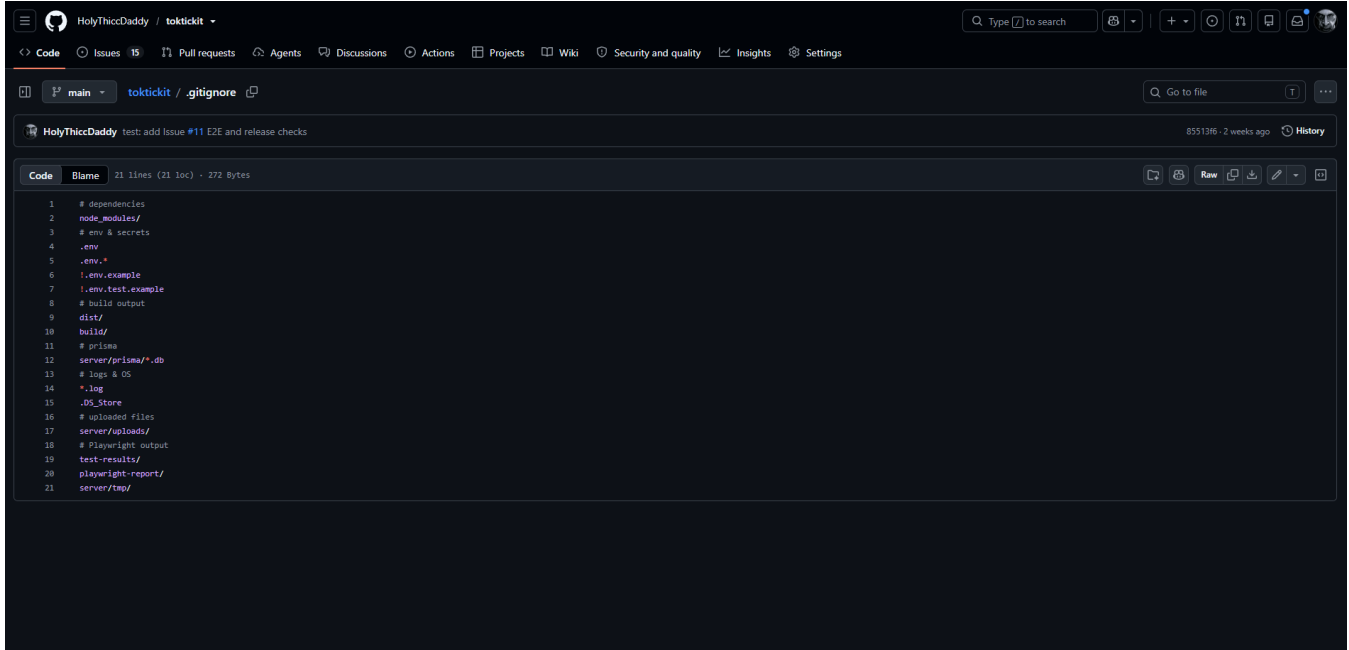
## Evidence 06 — Rendered README

```

1  # tokickit - Lab 3 Authentication & Ticketing
2
3  TokickIT is a full-stack IT service-desk application. Lab 3 replaces Lab 2's temporary Development Requester selector with authenticated sessions and adds the complete requester, IT Staff, and Administrator workflows. Requesters manage only their own tickets and attachments.
4
5  # Scope and Technology
6
7  **Frontend** React 18, TypeScript, Vite, Bootstrap 5
8  **Backend** Node.js, Express, TypeScript, Prisma ORM
9  **Database** PostgreSQL
10 **Testing** Vitest, Supertest, React Testing Library, Playwright
11 **Architecture** Monorepo with 'client/', 'server/', and repository-root E2E specs
12
13 The released application includes the approved Lab 3 queue, ticket detail, communication, and user-management workflow described in 'docs/lab-03/'.
14
15 # Repository structure
16
17 ```text
18 tokickit/
19 ├── client/           # React UI, component tests, and Playwright Abandonmy
20 ├── server/          # Express API, Prisma ORM, and API tests
21 ├── e2e/             # Playwright journeys and deterministic setup/teardown
22 ├── docs/lab-02/      # Lab 2 contract, test plan, AI log, and review record
23 ├── docs/lab-03/      # Lab 3 contract, test plan, AI log, and review record
24 ├── artifacts/lab-02/  # Lab 2 test output and resumable/visual evidence
25 ├── artifacts/lab-03/  # Lab 3 resumable/visual evidence
26 ├── output/           # Final submission reports
27 ...
28
29 # Prerequisites
30
31 Node.js "20 or higher" (required by the current Playwright dependency)
32 PostgreSQL running locally on port "5432"
33 A PostgreSQL role that can create databases
34
35 # Installation and environment
36
37 ```bash
38 git clone https://github.com/HolyDaddy/tokickit.git
39 cd tokickit
40 cd server
41 npm install
42 copy .env.example .env
43 cd ../client
44 npm install
45
46 Set 'server.env' to a local development database, for example:
47
48 ```env
49 DATABASE_URL="postgresql://tokickit:tokickit@localhost:5432/tokickit-test-db"
50 PORT=3000
51 TOKICKIT_API_URL=http://localhost
52
53
54 Create the dedicated test database before running destructive suites:
55
56 ```bash
57 createdb -U tokickit tokickit_test
58 cd server
59 copy .env.test.example .env.test
60 npm run migrate:dev
61 npm run prisma:seed
62 ...
63
64 'server.env.test' is local-only and ignored by Git. Test setup requires a database name ending in '_test' and writes attachments beneath a test-only temporary directory.
65
66 # Run the application
67
68 ```bash
69 # terminal 1
70 cd server
71 npm run dev
72
73 # terminal 2
74 cd client
75 npm run dev
76
77 The API is available at "http://localhost:3000" and Vite serves the UI at "http://localhost:5173".
78
79 # Verification commands
80
81 Run the smallest relevant suite first, then the full release checks:
82
83 ```bash
84 cd server
85 npm test
86 npm run build
87
88 cd ../client
89 npm test
90 npm run build
91 npm run playwright test
92
93 The Playwright command starts isolated services, resets deterministic fixtures in 'tokickit_test', checks requester, staff, and administrator journeys at desktop/tablet/mobile sizes, and writes JSOR results plus PMO evidence under the 'artifacts/' directories.
94
95 # API surface
96
97 - GET /api/health - service health
98 - POST /api/auth/login - create an authenticated session
99 - POST /api/auth/logout - revoke the current session
100 - GET /api/auth/me - return the current user summary
101 - GET /api/auth/reset - issue a CSRF token for state-changing requests
102 - POST /api/auth/change-password - simulate a first-time password change
103 - GET /api/categories - active ticket categories
104 - GET /api/queue/status - active related systems
105 - POST /api/tickets - create a 'new' ticket owned by the signed-in Requester
106 - GET /api/tickets - session-scoped search, filtering, sorting, and pagination
107 - GET /api/tickets/{id} - read-only ticket detail with public conversation
108 - POST /api/tickets/{id}/attachments - add a validated attachment to an owned ticket
109 - GET /api/attachments/{id}/download - download an active owned attachment
110 - GET /api/attachments/{id} - retrieve an owned attachment with a reason
111 - GET /api/tickets/{id}/comments - read or append public comments on an owned ticket
112 - POST /api/tickets/{id}/requester-resolution - record the Requester's resolution indication
113 - GET /api/staff/tickets/{id}/... - staff/queue assignment, IT priority, and status operations
114 - GET /api/tickets/{id}/notes - internal notes for IT Staff and Administrators
115 - GET /api/patch /api/patch/users... - Administrator user listing and safe account management
116
117 Protected identity is derived only from the server-side 'tokickit_session' cookie; client-supplied requester, owner, author, and role values are ignored. State-changing requests require the server-issued CSRF token. Attachment uploads enforce allowed type, extension,
118
119 # Lab 2 documentation
120
121 [Sprint specification](docs/lab-02/sprint-specification.md)
122 [NEXT API contract](docs/lab-02/api-contract.md)
123 [UI specification](docs/lab-02/ui-specification.md)
124 [Test plan and results](docs/lab-02/test-plan.md)
125 [AI use and reflection](docs/lab-02/ai-use.md)
126 [Peer-review record](docs/lab-02/peer-review.md)
127 [Lab 2 visual evidence](artifacts/lab-02/screenshots/)
128
129 # Lab 3 documentation
130
131 [Sprint specification](docs/lab-03/sprint-specification.md)
132 [NEXT API contract](docs/lab-03/api-contract.md)
133 [UI specification](docs/lab-03/ui-specification.md)
134 [Test plan and results](docs/lab-03/test-plan.md)
135 [AI use and reflection](docs/lab-03/ai-use.md)
136 [Peer-review record](docs/lab-03/peer-review.md)
137 [Final submission report](docs/lab-03/final-submission-report.pdf)
138
139 The core Lab 3 implementation was integrated into 'main' by release PR #22 ('a5f0b07'). The README/documentation update was prompted by PR #22 ('3d63f77'), and the final evidence/report update was prompted by PR #22 ('3d63f77').
140
141 Lab 3 was integrated into 'main' by Final Release PR #66 ('970c3c7') after peer-reviewed implementation PRs #42-#45 were merged into 'lab-staging'.
```

Original local capture retained under artifacts/lab-03/screenshots/report/06\_readme.png.

## Evidence 07 — Repository .gitignore



The screenshot shows a GitHub repository for 'HolyThiccDaddy / toktickit'. The file '.gitignore' is selected, showing 21 lines of code. The repository has 15 issues, 0 pull requests, 0 agents, 0 discussions, 0 actions, 0 projects, 0 wiki, 0 security and quality, 0 insights, and 0 settings. The file content is as follows:

```
1 # dependencies
2 node_modules/
3 # env & secrets
4 .env
5 .env.*
6 !env.example
7 !env.test.example
8 # build output
9 dist/
10 build/
11 # prisma
12 server/prisma/*.db
13 # logs & OS
14 *.log
15 .DS_Store
16 # uploaded files
17 server/uploads/
18 # Playwright output
19 test-results/
20 playwright-report/
21 server/tmp/
```

Original local capture retained under artifacts/lab-03/screenshots/report/07\_gitignore.png.

## Evidence 08 — Repository tree and README

The screenshot displays the GitHub repository page for **TokTickIT — Lab 3 Authenticated IT Ticketing**. The repository is owned by **HyPhenSecurity** and is a public repository. The README file is selected, showing the project's purpose, scope, and installation instructions.

**Repository tree:**

- main (23 branches)
- addition: Merge pull request #107 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #106 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #105 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #104 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #103 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #102 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #101 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #100 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #99 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #98 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #97 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #96 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #95 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #94 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #93 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #92 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #91 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #90 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #89 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #88 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #87 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #86 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #85 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #84 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #83 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #82 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #81 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #80 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #79 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #78 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #77 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #76 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #75 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #74 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #73 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #72 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #71 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #70 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #69 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #68 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #67 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #66 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #65 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #64 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #63 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #62 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #61 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #60 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #59 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #58 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #57 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #56 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #55 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #54 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #53 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #52 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #51 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #50 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #49 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #48 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #47 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #46 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #45 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #44 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #43 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #42 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #41 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #40 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #39 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #38 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #37 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #36 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #35 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #34 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #33 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #32 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #31 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #30 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #29 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #28 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #27 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #26 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #25 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #24 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #23 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #22 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #21 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #20 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #19 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #18 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #17 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #16 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #15 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #14 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #13 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #12 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #11 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #10 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #9 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #8 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #7 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #6 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #5 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #4 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #3 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #2 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago
- addition: Merge pull request #1 from HyPhenSecurity/feature/08-evidence-08 (v0.0.0) 7 months ago

**README:**

### TokTickIT — Lab 3 Authenticated IT Ticketing

TokTickIT is a full-stack IT service desk application. Lab 3 replaces Lab 2's temporary, development-requester-selector with authenticated sessions and adds the complete Requester, IT Staff, and Administrator workflows. Requesters manage only their own tickets and attachments, IT Staff and Administrators operate the shared queue, and Administrators manage user safety.

#### Scope and technology

- Frontend: React 18, TypeScript, Vite, Bootstrap 5
- Backend: Node.js, Express, TypeScript, Prisma ORM
- Database: PostgreSQL
- Testing: Vitest, Supertest, Jest, Playwright, Cypress
- Architecture: Monorepo with client/, server/, and repository/ root E2E specs

The released application includes the approved Lab 3 queue, ticket detail, communication, and user management workflows described in [docs/lab-03](#).

#### Repository structure

```
toktickit/
├── client/
│   ├── src/
│   │   ├── components/
│   │   ├── hooks/
│   │   ├── pages/
│   │   ├── services/
│   │   ├── stores/
│   │   ├── types/
│   │   ├── utils/
│   │   └── vite.config.ts
│   └── index.html
├── server/
│   ├── src/
│   │   ├── controllers/
│   │   ├── middleware/
│   │   ├── models/
│   │   ├── routes/
│   │   ├── services/
│   │   ├── types/
│   │   └── utils/
│   └── package.json
└── repository/
    ├── src/
    │   ├── adapters/
    │   ├── entities/
    │   ├── interfaces/
    │   ├── repositories/
    │   ├── services/
    │   ├── types/
    │   └── utils/
    └── package.json
```

#### Prerequisites

- Node.js 20 or higher (required by the current Playwright dependency)
- PostgreSQL running locally on port 5432
- A PostgreSQL role that can create databases

#### Installation and environment

```
git clone https://github.com/HyPhenSecurity/toktickit.git
cd toktickit
npm install
npm run example:dev
cd ./client
npm install
```

Set `server/ssl` to a local development database, for example:

```
export DB_URL="postgresql://toktickit:toktickit@localhost:5432/toktickit?ssl=1"
export DB_NAME="toktickit"
export DB_SCHEMA="toktickit"
```

Create the dedicated test database before running destructive suites:

```
createdb -U toktickit toktickit_test
cd server
npm run test:example:dev
npm run migrate:dev
npm run migrate:prod
```

`server/ssl.test` is local-only and ignored by Git. Test setup requires a database name ending in `_test` and writes attachments beneath a test-only temporary directory.

#### Run the application

```
cd repository
cd server
npm run dev
cd client
npm run dev
```

The API is available at <http://localhost:3000> and the server the UI at <http://localhost:3001>.

#### Verification commands

Run the smallest relevant suite first, then the full release check:

```
cd server
npm run test
npm run build
cd ./client
npm run dev
npm run build
npm run migrate:prod
npm run playwright:prod
```

The Playwright command starts isolated sessions, reads deterministic fixtures in `toktickit_test`, checks requests, staff, and administrator journeys at desktop/tablet/mobile sizes, and writes JSON results plus PNG evidence under the `toktickit/` directory.

#### API surface

- `GET /api/health` — service health
- `POST /api/auth/login` — create an authenticated session
- `POST /api/auth/logout` — revoke the current session
- `GET /api/auth/me` — return the current user summary
- `POST /api/auth/reset` — issue a CSRF token for state-changing requests
- `POST /api/auth/change-password` — complete a first login password change
- `GET /api/auth/session` — active ticket categories
- `POST /api/auth/session` — create a new ticket owned by the signed-in Requester
- `GET /api/auth/session` — session request status, filtering, sorting, and pagination
- `GET /api/auth/session` — internal audit only ticket detail with public conversation
- `POST /api/auth/session/attachments` — add a validated attachment to an owned ticket
- `GET /api/auth/session/attachments` — download an active owned attachment
- `DELETE /api/auth/session/attachments` — soft-remove an owned attachment with a reason
- `GET /api/auth/session/comments` — read or append public comments on an owned ticket
- `POST /api/auth/session/request-resolution` — record the Requester's resolution indication
- `GET /api/auth/session/status` — staff/admin assignment, if priority and status operations
- `GET /api/auth/session/status` — internal notes for IT Staff and Administrators
- `GET /api/auth/session/status` — Administrative user listing and role-account management

Protected identity is derived only from the server-side `toktickit_session` cookie; client-supplied requests, owners, authors, and role values are ignored. State-changing requests require the server-issued CSRF token. Attachment uploads enforce allowed type, extension, image/bytes, size, count, ownership, and corresponding cleanup rules.

#### Lab 2 documentation

- Secret specifications
- UI specifications
- Test plan and results
- API use and reflection
- Free release record
- Lab 2 final evidence

#### Lab 3 documentation

- Secret specifications
- UI specifications
- Test plan and results
- API use and reflection
- Free release record
- Final submission report source
- Final submission report E2E

The core Lab 2 implementation was integrated into `main` by release PR #27 (446603). The README/documentation update was promoted by PR #29 (446607), and the final evidence/report update was promoted by PR #32 (446608). Lab 3 was integrated into `main` by Final Release PR #46 (525623) after peer-reviewed implementation PRs #40-#45 were merged into `main` (346166).

Original local capture retained under artifacts/lab-03/screenshots/report/08\_repository\_tree\_readme.png.

## Evidence 12 — PR #40 changed files

HolyThiccDaddy / toktickit

< Code Issues Pull requests Agents Discussions Actions Projects Wiki Security and quality Insights Settings

Issue #35: Sprint 3 engineering contract #40

Merged osizk merged 4 commits into lab3-staging from feature/lab3-contract last week

Conversation Commits Checks Files changed

All commits

0 / 13 viewed Submit comments

artifacts/lab-03/screenshots/authentication/.gitkeep  
@@ -0,0 +1 @@  
1 +

artifacts/lab-03/screenshots/staff-queue/.gitkeep  
@@ -0,0 +1 @@  
1 +

artifacts/lab-03/screenshots/staff-ticket-detail/.gitkeep  
@@ -0,0 +1 @@  
1 +

artifacts/lab-03/screenshots/user-management/.gitkeep  
@@ -0,0 +1 @@  
1 +

client/tests/lab-03/.gitkeep  
@@ -0,0 +1 @@  
1 +

docs/lab-03/ai-use.md  
@@ -0,0 +1,20 @@  
1 + # Lab 3 AI Use and Reflection  
16 + | B | Audit release evidence | Check source, logs, screenshots, and final report consistency | To be recorded before release PR |  
17 +  
18 + ## Reflection  
19 +  
20 + AI output is treated as a draft. Contract decisions remain subject to the Lab 3 handout and peer review. Security-sensitive behavior is accepted only when executable tests and real terminal evidence confirm it;  
suggestions that weaken server-side authorization, expose credentials, or invent unsupported requirements are rejected.

docs/lab-03/api-spec.md  
@@ -0,0 +1,136 @@  
1 + # Lab 3 API Specification  
2 +  
3 + Status: Draft contract for peer review before implementation  
4 +  
5 + ## Conventions  
6 +  
7 + - Base path: /api.  
8 + - 3500 requests and responses use UTF-8 and the standard envelope { data: ... } on success.  
9 + - Failures use { error: { code, message, fieldErrors } }. Codes are stable; messages never expose secrets, SQL, filesystem paths, storage keys, or unnecessary account details.  
10 + - Browser authentication uses an opaque HttpOnly cookie named "toktickit-session"; the server resolves the user and role from the session.  
11 + - GET endpoints are safe. Every authenticated POST, PATCH, and DELETE requires the CSRF token returned by GET "/api/auth/csrf" in "X-CSRF-Token"; the public login request is the exception.  
12 + - "401 UNAUTHENTICATED" means no valid session; "403 FORBIDDEN" means a valid session lacks the role or ownership; protected ticket resources may return "404 NOT\_FOUND" for both missing and unauthorized records.  
13 + - Validation is "400 VALIDATION\_ERROR"; duplicate or invalid state is "409 CONFLICT"; unexpected failures are "500 INTERNAL\_ERROR".  
14 + - A session with "mustChangePassword=true" is authenticated but restricted. Until the password is changed, it may call only "/auth/me", "/auth/csrf", "/auth/change-password", and "/auth/logout". Every other protected endpoint returns "403 PASSWORD\_CHANGE\_REQUIRED" without performing a mutation.

Original local capture retained under artifacts/lab-03/screenshots/report/12\_pr40\_files\_changed.png.

## Evidence 13 — PR #40 commits

The screenshot shows the GitHub web interface for a repository named 'HolyThiccDaddy / toktickit'. The top navigation bar includes links for Code, Issues (15), Pull requests, Agents, Discussions, Actions, Projects, Wiki, Security and quality, Insights, and Settings. A search bar is located on the right. The main content area displays 'Issue #35: Sprint 3 engineering contract #40' with a 'Code' button. Below the issue title, it states 'osix merged 4 commits into 'lab3-staging' from 'feature/lab3-contract' last week'. There are tabs for Conversation (11), Commits (4), Checks (0), and Files changed (13). A green progress bar indicates 4/4 commits. Below the tabs, a section titled 'Commits on Sep 12, 2026' lists four commits:

| Commit Message                               | Commit Hash | Author                             |
|----------------------------------------------|-------------|------------------------------------|
| docs: add Lab 3 engineering contract         | 2a5513a     | HolyThiccDaddy committed last week |
| docs: link Lab 3 contract review PR          | 4e988a9     | HolyThiccDaddy committed last week |
| docs: address Lab 3 contract review feedback | 887c296     | HolyThiccDaddy committed last week |
| docs: complete Lab 3 contract review updates | 24e7857     | HolyThiccDaddy committed last week |

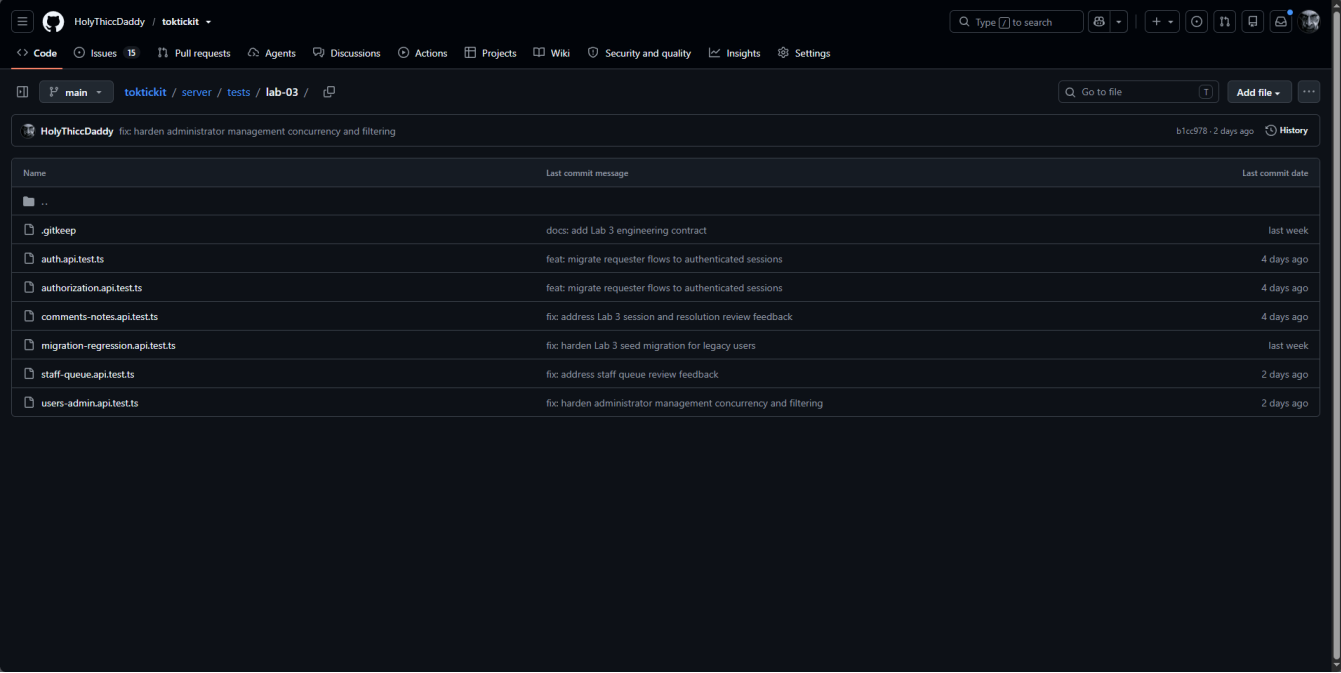
Original local capture retained under artifacts/lab-03/screenshots/report/13\_pr40\_commits.png.

## Evidence 14 — Rendered tests plan and results

```
HolyThiccDaddy / toktickit +
Code Issues 15 Pull requests Agents Discussions Actions Projects Wiki Security and quality Insights Settings
feature/lab3-final... - toktickit / docs / lab-03 / tests.md
HolyThiccDaddy docs: finalize Lab 3 main release evidence 644db81 - 1 hour ago History
Preview Code Blame 81 lines (65 loc) - 10.8 KB
1 # Lab 3 Test Plan and Results
2
3 Status: Final Issue #22 implementation and responsive release evidence were verified from 'main' after Final Release PR #26 merged as commit '92fac36'.
4
5 ## 1. Testing strategy
6
7 Use unit, API/integration, UI component, UI style, responsive, security/authorization, migration/regression, and Playwright E2E layers. Every acceptance criterion in specification.md is mapped below. Test fixtures use server/.env.test, a database name ending in _test,
8
9 ## 2. Traceability matrix
10
11 | ID | Acceptance criterion | Layer | Planned test file / evidence |
12 |---|---|---|---|
13 | T-01 | AC-01 valid login and role shell | API + UI + E2E | server/tests/lab-03/auth.api.test.ts; client/tests/lab-03/login.test.ts; e2e/lab-03/authentication.spec.ts |
14 | T-02 | AC-02 invalid/inactive login safety | API + UI + E2E | auth.api.test.ts; login.test.ts; authentication.spec.ts |
15 | T-03 | AC-03 first-login password gate and direct API blocking | API + UI + security + E2E | auth.api.test.ts; authorization.api.test.ts; client/tests/lab-03/ChangePassword.test.ts; authentication.spec.ts |
16 | T-04 | AC-04 logout/expiry protection | API + security + E2E | auth.api.test.ts; authorization.api.test.ts; authentication.spec.ts |
17 | T-05 | AC-05 requester Lab 2 regression/ownership | API + UI + migration + E2E | server/tests/lab-02/create-ticket.api.test.ts; my-tickets.api.test.ts; ticket-detail.api.test.ts; e2e/lab-02/multi-requester-isolation.spec.ts |
18 | T-06 | AC-06 requester public comment/resolution indication and ownership | API + UI + security + E2E | server/tests/lab-03/comments-notes.api.test.ts; client/tests/lab-03/RequesterTicketDetail.test.ts; e2e/lab-03/requester-regression.spec.ts |
19 | T-07 | AC-07 shared queue search/filters/sort/pagination/responsive states | API + UI + style + responsive + E2E | server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketQueue.test.ts; e2e/lab-03/requester-regression.spec.ts |
20 | T-08 | AC-08 claim/reassign/IT Priority enum/complete status matrix authorization | API + UI + security + E2E | server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketDetail.test.ts; e2e/lab-03/staff-ticket-flow.spec.ts |
21 | T-09 | AC-09 public vs internal visibility and append-only validation | API + UI + security + E2E | server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketDetail.test.ts; e2e/lab-03/staff-ticket-flow.spec.ts |
22 | T-10 | AC-10 user management and activation state | API + UI + E2E | server/tests/lab-03/users-admin.api.test.ts; client/tests/lab-03/UserManagement.test.ts; e2e/lab-03/user-administration.spec.ts |
23 | T-11 | AC-11 self-deactivation/self-role-change, last-admin, and deactivated-user safety | API + security + UI | users-admin.api.test.ts; authorization.api.test.ts; UserManagement.test.ts |
24 | T-12 | AC-12 migration/data preservation/depotment seed | Migration + regression | server/tests/lab-03/migration-regression.api.test.ts; migration terminal evidence |
25 | T-13 | AC-13 forged identity/wrong-role rejection | API + security | server/tests/lab-03/authorization.api.test.ts |
26 | T-14 | AC-14 responsive no-overflow behavior | UI style + responsive + E2E | client/tests/lab-03/ResponsiveLayout.test.ts; all Lab 3 E2E projects; artifacts/lab-03/screenshots |
27 | T-15 | AC-15 release gates | Regression + build + E2E | server/client test and build logs; e2e results; release checklist |
28
29 T-03 must call a protected Ticket and Admin endpoint directly while 'mustChangePassword=true' and assert '403 PASSWORD_CHANGE_REQUIRED', including an uppercase or mixed-case path variant; only '/auth/me', '/auth/csr', '/auth/change-password', and '/auth/logout' remain
30
31 T-06 must exercise a Requester-owned Ticket Detail, successful Public Comment submission, successful Problem Appears Resolved indication, rejection for a different Requester, and an assertion that the indication does not change formal status to RESOLVED or CLOSED.
32
33 ## 3. Planned server suites
34
35 - server/tests/lab-03/auth.api.test.ts: valid/invalid login, inactive-account response after password verification, logout, session, CSRF, password policy, first-login gate, expiry, and response schema/status assertions.
36 - server/tests/lab-03/authorization.api.test.ts: session-derived identity, role boundaries, ownership, deactivated users, must-change-password gate, and safe errors.
37 - server/tests/lab-03/staff-queue.api.test.ts: queue search/filter/sort/pagination, invalid query behavior, claim race, reassignment to active IT Staff/Admin, IT Priority enum validation, status confirmation, staff detail, and response metadata.
38 - server/tests/lab-03/comments-notes.api.test.ts: Requester/Staff/Admin public comments, internal notes, visibility, append-only behavior, length/whitespace validation, and ownership.
39 - server/tests/lab-03/users-admin.api.test.ts: admin CRUD-like operations, role validation, activation on create/edit, initial password, self-deactivation/self-role-change, duplicate email, and last-admin guards.
40 - server/tests/lab-03/migration-regression.api.test.ts: preserved Lab 2 records, data-model relationships, Requested-to-IT Priority copy, and deterministic repeated seed.
41
42 ## 4. Planned client and E2E suites
43
44 - client/tests/lab-03/Login.test.tsx, ChangePassword.test.tsx, and RequesterTicketDetail.test.tsx.
45 - client/tests/lab-03/StaffTicketQueue.test.tsx, StaffTicketDetail.test.tsx, UserManagement.test.tsx, and ResponsiveLayout.test.tsx.
46 - e2e/lab-03/authentication.spec.ts, requester-regression.spec.ts, staff-ticket-flow.spec.ts, and user-administration.spec.ts.
47 - Screenshots are captured from real runs under artifacts/lab-03/screenshots/authentication, staff-queue, staff-ticket-detail, and user-management.
48
49 ## 5. Release checklist
50
51 - [x] All planned tests implemented and passing.
52 - [x] Server and client builds pass.
53 - [x] Playwright desktop, tablet, and mobile projects pass.
54 - [x] Migration and repeated seed pass twice against isolated_test database.
55 - [x] Security checks cover missing session, forged identity, wrong role, ownership, CSRF, inactive user, and Administrator safety guards.
56 - [x] First-login sessions cannot bypass the password gate through direct API calls.
57 - [x] API response schemas, status codes, data-model decisions, and IT Priority enum are covered by tests.
58 - [x] Terminal logs and visual evidence are captured and linked.
59 - [x] Results below are updated only with commands actually run.
60
61 ## 6. Executed results
62
63 - Issue #26 authentication API: 'server/tests/lab-03/auth.api.test.ts' - 10 tests passed, including generic inactive-account failures, case-insensitive first-login API-gate bypass checks, malformed JSON handling, expired sessions, and CSRF expiry.
64 - Issue #26 migration and deterministic-fixture regression: 'server/tests/lab-03/migration-regression.api.test.ts' - 7 tests passed, including canonical ownership links, repeated-seed equality, preservation of Lab 2 reference/counter/ticket/attachment rows, a legacy re
65 - Issue #22 requester authorization and communication regression: 'server/tests/lab-03/authorization.api.test.ts' plus 'server/tests/lab-03/comments-notes.api.test.ts' - 11 tests passed, covering session-derived identity, forged-header rejection, role/ownership boundar
66 - Issue #22 staff queue and ticket operations: 'server/tests/lab-03/staff-queue.api.test.ts' - 35 tests passed, covering shared queue queries, invalid-query safety, atomic claim, eligible reassignment, IT Priority, every valid BR-08 transition, required-confirmation fi
67 - Issue #22 staff UI components: 'client/tests/lab-03/StaffTicketQueue.test.tsx' and 'StaffTicketDetail.test.tsx' - 4 tests passed, covering queue states/filters/page size/result count, mobile card sorting controls, retryable validation failure, claim, priority/status
68 - Issue #22 staff E2E: 'client npm run test:e2e -- -c2e/lab-03/staff-ticket-flow.spec.ts' - 2 tests passed on the desktop project, covering requester-created work entering the shared staff queue, opening staff detail, and mobile card sorting controls at 375px.
69 - Issue #22 Administrator API: 'server/tests/lab-03/users-admin.api.test.ts' - 4 tests passed, covering safe listing and role boundaries, create/edit/activation/initial-password reset, duplicate email, invalid role, inactive login, self-safety rules, and concurrent la
70 - Issue #22 Administrator UI: 'client/tests/lab-03/UserManagement.test.tsx' - 6 tests passed, covering list/search controls, filtered mutation refresh, stale-search response protection, partial edit payloads, filter changes during a pending mutation, create/edit/activ
71 - Issue #22 E2E: 'client npm run test:e2e -- -c2e/lab-03 -project=desktop' - 6 tests passed, covering authentication, requester communication, staff queue/detail, and Administrator user management.
72 - Issue #22 responsive E2E: 'client npm run test:e2e -- -c2e/lab-03/responsive.visual.spec.ts -project=tablet -project=mobile' - 2 tests passed; each project captured authentication, User Management, Staff Queue, and Staff Ticket Detail evidence.
73 - Full server regression: 'server npm test' - 15 test files, 92 tests passed.
74 - Full client regression: 'client npm vitest run --maxWorkers=1 --minWorkers=1' - 15 test files, 50 tests passed, including Administrator User Management, mobile staff queue sorting, and authenticated requester recovery. The serial run is the verified result.
75 - Authenticated requester E2E baseline: the retained evidence records 6 tests passed across desktop, tablet, and mobile projects, including public comment/resolution flow; a later full rerun retained six functional passes but its desktop visual capture hit an existing
76 - Server and client production builds passed after the Issue #22 implementation.
77 - 'npm prism validate' passed.
78 - 'npm prism migrate deploy' applied the two Lab 3 migrations to the isolated 'toktickit-test' database.
79 - Final integration: PR #26 merged the reviewed Lab 3 branch into 'main' as commit '92fac36'; the release-candidate verification above was run against the same merged source.
80
81 Issue #22 evidence is stored under 'artifacts/lab-03/screenshots/authentication,user-management,staff-queue,staff-ticket-detail' with desktop, tablet, and mobile captures from the real Playwright runs above.
```

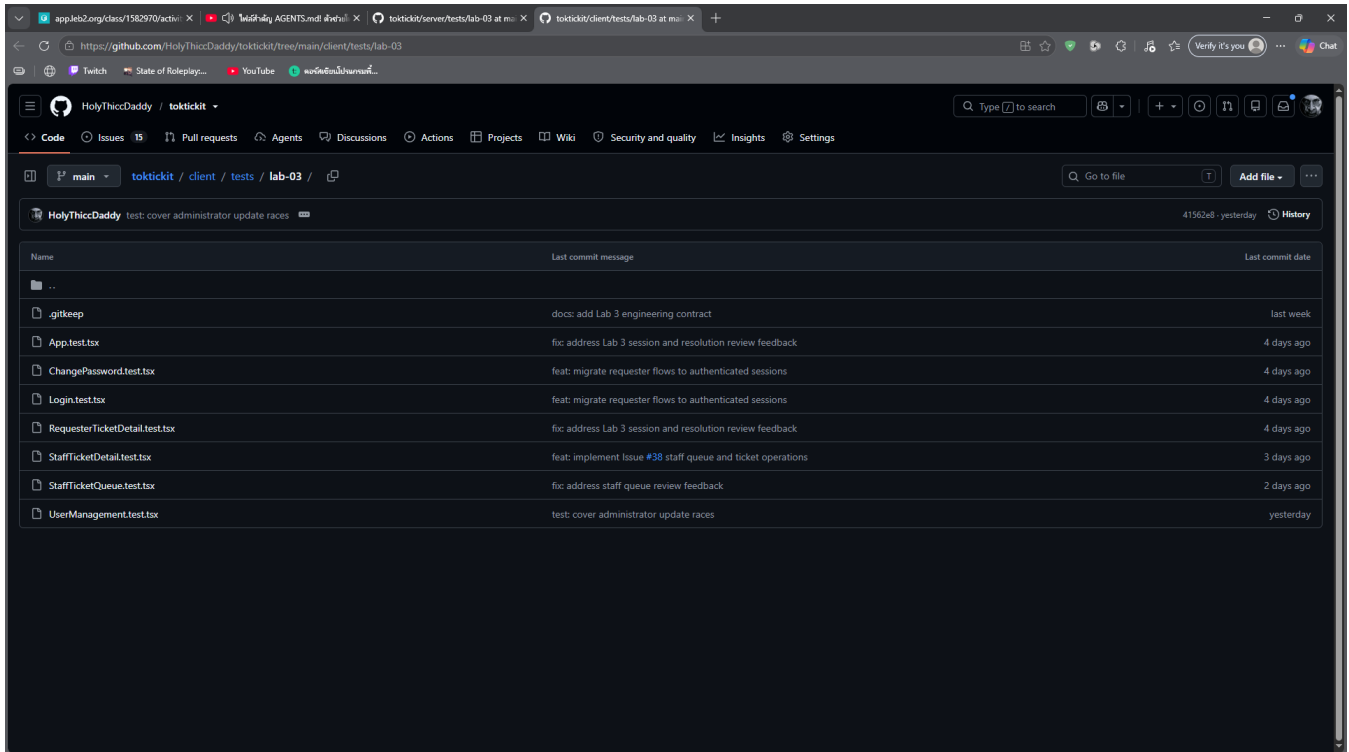
Original local capture retained under artifacts/lab-03/screenshots/report/14\_tests\_plan\_results.png.

Evidence 15 — Server Lab 3 test tree



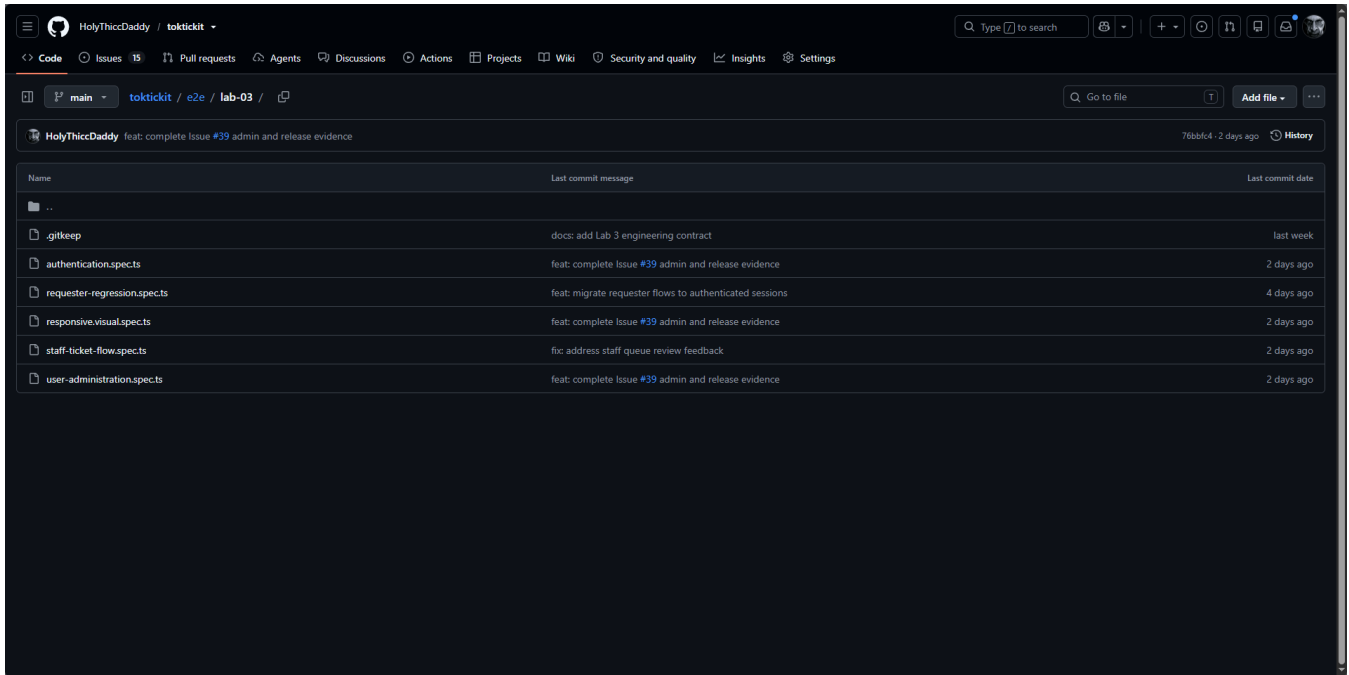
Original local capture retained under artifacts/lab-03/screenshots/report/15\_server\_lab3\_test\_tree.png.

## Evidence 16 — Client Lab 3 test tree



Original local capture retained under artifacts/lab-03/screenshots/report/16\_client\_lab3\_test\_tree.png.

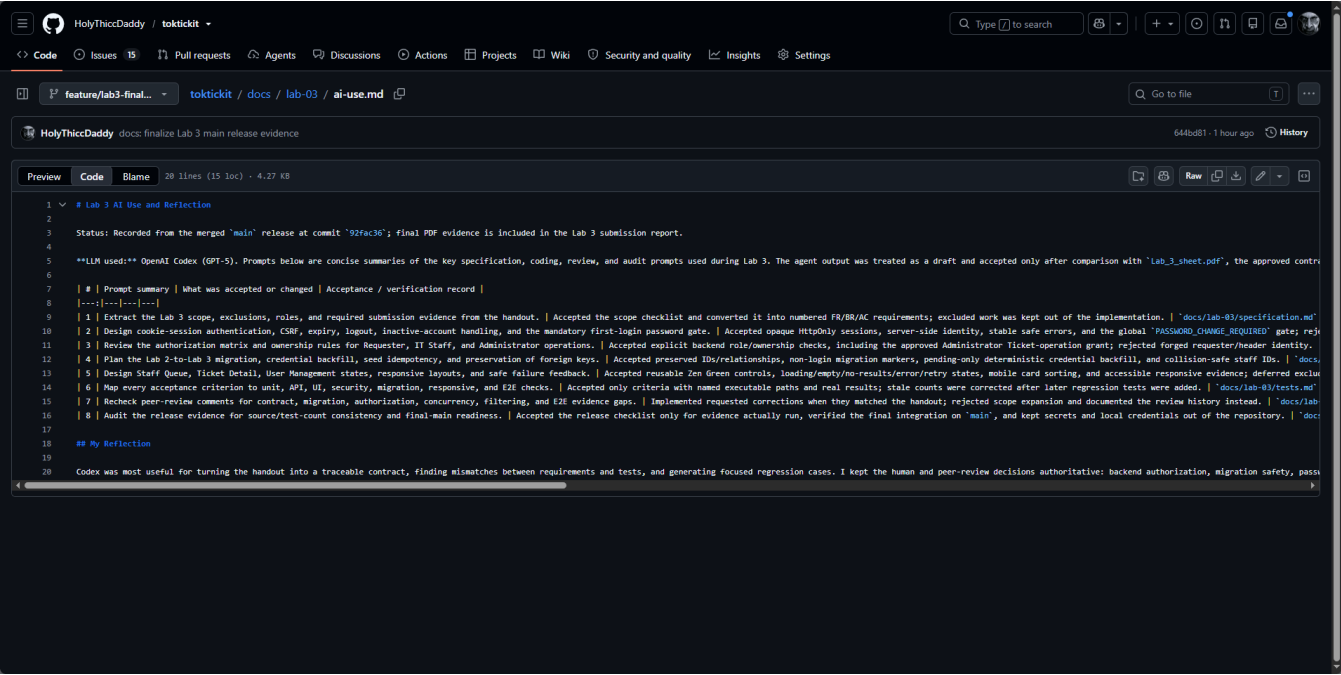
## Evidence 17 — Playwright Lab 3 test tree



| Name                        | Last commit message                                     | Last commit date |
|-----------------------------|---------------------------------------------------------|------------------|
| ..                          |                                                         |                  |
| .gitkeep                    | docs: add Lab 3 engineering contract                    | last week        |
| authentication.spects       | feat: complete Issue #39 admin and release evidence     | 2 days ago       |
| requester-regression.spects | feat: migrate requester flows to authenticated sessions | 4 days ago       |
| responsive.visual.spects    | feat: complete Issue #39 admin and release evidence     | 2 days ago       |
| staff-ticket-flow.spects    | fix: address staff queue review feedback                | 2 days ago       |
| user-administration.spects  | feat: complete Issue #39 admin and release evidence     | 2 days ago       |

Original local capture retained under artifacts/lab-03/screenshots/report/17\_e2e\_lab3\_test\_tree.png.

Evidence 18 — Rendered AI-use record



Original local capture retained under artifacts/lab-03/screenshots/report18\_ai\_use.png.

## Evidence 19 — Release branch/commit context

```
● PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\capture-main
● PS D:\Downloads\toktickit\capture-main> git log -1 --oneline --decorate
e7998d3 (HEAD, origin/main) Merge pull request #47 from HolyThiccDaddy/feature/lab3-final-evidence
● PS D:\Downloads\toktickit\capture-main> cd D:\Downloads\toktickit\capture-main\server
PS D:\Downloads\toktickit\capture-main\server> npm test

> toktickit-server@1.0.0 test
> vitest run

RUN v2.1.9 D:/Downloads/toktickit/capture-main/server

✓ tests/lab-01/categories.test.ts (1)
✓ tests/lab-01/health.test.ts (1)
✓ tests/lab-02/attachment-validator.unit.test.ts (2)
✓ tests/lab-02/create-ticket.api.test.ts (4) 792ms
✓ tests/lab-02/my-tickets.api.test.ts (5) 652ms
✓ tests/lab-02/requesters.api.test.ts (1)
✓ tests/lab-02/seed.test.ts (6) 716ms
✓ tests/lab-02/ticket-detail.api.test.ts (4) 581ms
✓ tests/lab-02/ticket-number.unit.test.ts (1)
✓ tests/lab-03/auth.api.test.ts (10) 1467ms
✓ tests/lab-03/authorization.api.test.ts (5) 1209ms
✓ tests/lab-03/comments-notes.api.test.ts (6) 1142ms
✓ tests/lab-03/migration-regression.api.test.ts (7) 1887ms
✓ tests/lab-03/staff-queue.api.test.ts (35) 9224ms
✓ tests/lab-03/users-admin.api.test.ts (4) 783ms

Test Files 15 passed (15)
Tests 92 passed (92)
Start at 18:39:17
Duration 28.76s (transform 252ms, setup 0ms, collect 4.96s, tests 18.70s, environment 3ms, prepare 1.77s)

● PS D:\Downloads\toktickit\capture-main\server> npm run build

> toktickit-server@1.0.0 build
> tsc

○ PS D:\Downloads\toktickit\capture-main\server> 
```

Original local capture retained under artifacts/lab-03/screenshots/report/19\_terminal\_branch\_status.png.

Evidence 20 — Zen Green token capture

```
PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\capture-main
PS D:\Downloads\toktickit\capture-main> Write-Host "=== Zen Green tokens ===" -ForegroundColor Cyan
=== Zen Green tokens ===
PS D:\Downloads\toktickit\capture-main> $lines = Get-Content .\docs\lab-02\ui-spec.md
PS D:\Downloads\toktickit\capture-main> $lines[0..24]
# Lab 2 Zen Green UI Specification

## 1. Overview & Design Philosophy
TokTickIT Lab 2 establishes the Zen Green Theme, a cohesive, professional, and accessible design system for enterprise IT service desks. All screens in Lab 2 and subsequent labs strictly reuse these component primitives, color tokens, and layout guidelines.

---

## 2. Color Tokens & Style Variables

| Token Name | Hex Value | Purpose / Usage |
| :--- | :--- | :--- |
| Primary Green | #00683C | Application header, primary action buttons, strong brand emphasis |
| Secondary Green | #007A46 | Active navigation tabs, focus rings, link hover states, subheaders |
| Pale Green | #EAF6EF | Selected rows, subtle section highlights, success container background |
| Page Background | #F5F7F6 | Default body/page canvas background |
| Surface / Card | #FFFFFF | Form cards, table containers, modal surfaces with subtle shadow |
| Text Primary | #1B2E24 | Dark charcoal-green for high-contrast, comfortable typography |
| Text Secondary | #52665A | Subtitles, helper text, table column headers, timestamps |
| Border Neutral | #D1DDD6 | Form input borders, table row dividers, card borders |
| Read-Only Shading | #EEF3F0 | Distinct background for non-editable input fields |
| Error Dark Red | #C53830 | Validation error text, field error borders, destructive actions |
| Error Background | #FFF5F5 | Inline error alert container background |
| Warning / Priority | #D69E2E | Medium/High priority badges, disclaimer banners |
| Urgent Priority | #E53E3E | Urgent priority badge |
| Low Priority | #319795 | Low priority badge |
```

Original local capture retained under artifacts/lab-03/screenshots/report/20\_terminal\_zen\_green\_tokens.png.

## Evidence 21 — Visual checklist capture

```
PS D:\Downloads\toktickit> Write-Host "=== Visual checklist ===" -ForegroundColor Cyan
• === Visual checklist ===
PS D:\Downloads\toktickit> $lines = Get-Content .\docs\lab-02\tests.md
• PS D:\Downloads\toktickit> $lines[87..95]
• ## 4. Responsive and Visual Checklist

- [x] **Zen Green Color Palette:** Header uses `#006B3C`, hover/active accents use `#0B7A46`, pale background uses `#EAF6EF`, body uses `#F5F7F6`.
- [x] **Typography & Spacing:** Standardized font scale, consistent form input heights (38px), labels positioned above controls with red asterisk markers.
- [x] **Button Hierarchy:** Primary green buttons for main actions, outline buttons for secondary actions, amber for warnings, red for destructive soft-removal.
- [x] **Desktop Viewport (>= 992px):** Centered layout with max-width container, multi-column form, full data table with sortable headers.
- [x] **Tablet Viewport (768px - 991px):** Two-column form layout, responsive table layout with horizontal scrolling or condensed columns.
- [x] **Mobile Viewport (< 768px):** Single-column stacked form, ticket cards replacing table rows, touch-friendly buttons >= 44px height.
- [x] **No Visual Defects:** No clipped text, overlapping elements, or unexpected horizontal scrollbars at any breakpoint.
• PS D:\Downloads\toktickit> █
```

Original local capture retained under artifacts/lab-03/screenshots/report/21\_terminal\_visual\_checklist.png.

## Evidence 22 — Responsive rules capture

```
PS D:\Downloads\toktickit> Write-Host "=== Responsive rules ===" -ForegroundColor Cyan
=== Responsive rules ===
• PS D:\Downloads\toktickit> $lines - Get-Content .\docs\lab-03\ui-spec.md
• PS D:\Downloads\toktickit> $lines[4..8]
• ## Design foundation

Continue the Lab 2 Zen Green visual language: dark neutral surfaces, green success/accent states, readable contrast, consistent focus rings, shared buttons/inputs/tables, and no horizontal overflow. Reuse existing tokens and components rather than introducing a second visual system.

Breakpoints: desktop at 992px and above, tablet 768px~991px, mobile below 768px. At every breakpoint content must reflow without clipping or overlap. Tables become stacked cards or scroll within the component only when required; the page itself must not overflow horizontally.
PS D:\Downloads\toktickit> █
```

Original local capture retained under artifacts/lab-03/screenshots/report/22\_terminal\_responsive\_rules.png.

## Evidence 23 — Accessibility capture

```
PS D:\Downloads\toktickit> Write-Host "=== Accessibility ===" -ForegroundColor Cyan
=== Accessibility ===
PS D:\Downloads\toktickit> $lines = Get-Content .\docs\lab-03\ui-spec.md
PS D:\Downloads\toktickit> $lines[45..51]
## Accessibility and responsive acceptance

- All controls have accessible labels and keyboard operation.
- Error text is associated with its field and announced in a live region when appropriate.
- Focus remains visible after dialogs, mutations, and responsive layout changes.
- At desktop, tablet, and mobile widths there is no page-level horizontal overflow, clipped text, or overlapping controls.
- Screenshot evidence must cover authentication, staff queue, staff detail, and user management at representative desktop/tablet/mobile widths.
PS D:\Downloads\toktickit> █
```

Original local capture retained under artifacts/lab-03/screenshots/report/23\_terminal\_accessibility.png.

## Evidence 24 — Server npm test capture

```
PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\.worktree-main-audit\server
PS D:\Downloads\toktickit\.worktree-main-audit\server> npm test

> toktickit-server@1.0.0 test
> vitest run

RUN v2.1.9 D:/Downloads/toktickit/.worktree-main-audit/server

✓ tests/lab-01/categories.test.ts (1)
✓ tests/lab-01/health.test.ts (1)
✓ tests/lab-02/attachment-validator.unit.test.ts (2)
✓ tests/lab-02/create-ticket.api.test.ts (4) 676ms
✓ tests/lab-02/my-tickets.api.test.ts (5) 671ms
✓ tests/lab-02/requesters.api.test.ts (1)
✓ tests/lab-02/seed.test.ts (6) 488ms
✓ tests/lab-02/ticket-detail.api.test.ts (4) 548ms
✓ tests/lab-02/ticket-number.unit.test.ts (1)
✓ tests/lab-03/auth.api.test.ts (10) 1274ms
✓ tests/lab-03/authorization.api.test.ts (5) 986ms
✓ tests/lab-03/comments-notes.api.test.ts (6) 1128ms
✓ tests/lab-03/migration-regression.api.test.ts (7) 2032ms
✓ tests/lab-03/staff-queue.api.test.ts (4) 1484ms
✓ tests/lab-03/staff-ticket-detail.api.test.ts (31) 9043ms
✓ tests/lab-03/users-admin.api.test.ts (4) 871ms

Test Files 16 passed (16)
Tests      92 passed (92)
Start at   19:53:18
Duration   30.14s (transform 240ms, setup 0ms, collect 5.34s, tests 19.44s, environment 3ms, prepare 1.82s)
```

Original local capture retained under artifacts/lab-03/screenshots/report/24\_terminal\_server\_tests.png.

## Evidence 25 — Client npm test capture

```
PS D:\Downloads\toktickit\.capture-main\client> npm test

> toktickit-client@1.0.0 test
> vitest run

RUN v2.1.9 D:/Downloads/toktickit/.capture-main/client

✓ tests/lab-01/App.test.tsx (3)
✓ tests/lab-02/api.test.tsx (7)
✓ tests/lab-02/CreateTicket.test.tsx (4) 512ms
✓ tests/lab-02/MyTickets.test.tsx (4) 463ms
✓ tests/lab-02/RequesterSelector.test.tsx (2)
✓ tests/lab-02/ResponsiveLayout.test.tsx (2)
✓ tests/lab-02/ThemeStyles.test.tsx (2)
✓ tests/lab-02/TicketDetail.test.tsx (6) 810ms
✓ tests/lab-03/App.test.tsx (2) 319ms
✓ tests/lab-03/ChangePassword.test.tsx (1)
✓ tests/lab-03/Login.test.tsx (3)
✓ tests/lab-03/RequesterTicketDetail.test.tsx (4) 353ms
✓ tests/lab-03/StaffTicketDetail.test.tsx (1) 625ms
✓ tests/lab-03/StaffTicketQueue.test.tsx (3) 713ms
✓ tests/lab-03/UserManagement.test.tsx (6) 1131ms

Test Files 15 passed (15)
Tests      50 passed (50)
Start at   23:17:43
Duration   7.22s (transform 590ms, setup 2.64s, collect 3.37s, tests 5.84s, environment 16.36s, prepare 2.67s)
```

Original local capture retained under artifacts/lab-03/screenshots/report/25\_terminal\_client\_tests.png.

## Evidence 26 — Server build capture

```
● PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\.capture-main
● PS D:\Downloads\toktickit\.capture-main> git log -1 --oneline --decorate
e7998d3 (HEAD, origin/main) Merge pull request #47 from HolyThiccDaddy/feature/lab3-final-evidence
● PS D:\Downloads\toktickit\.capture-main> cd D:\Downloads\toktickit\.capture-main\server
PS D:\Downloads\toktickit\.capture-main\server> npm test

> toktickit-server@1.0.0 test
> vitest run

RUN v2.1.9 D:/Downloads/toktickit/.capture-main/server

✓ tests/lab-01/categories.test.ts (1)
✓ tests/lab-01/health.test.ts (1)
✓ tests/lab-02/attachment-validator.unit.test.ts (2)
✓ tests/lab-02/create-ticket.api.test.ts (4) 792ms
✓ tests/lab-02/my-tickets.api.test.ts (5) 652ms
✓ tests/lab-02/requesters.api.test.ts (1)
✓ tests/lab-02/seed.test.ts (6) 716ms
✓ tests/lab-02/ticket-detail.api.test.ts (4) 581ms
✓ tests/lab-02/ticket-number.unit.test.ts (1)
✓ tests/lab-03/auth.api.test.ts (10) 1467ms
✓ tests/lab-03/authorization.api.test.ts (5) 1209ms
✓ tests/lab-03/comments-notes.api.test.ts (6) 1142ms
✓ tests/lab-03/migration-regression.api.test.ts (7) 1887ms
✓ tests/lab-03/staff-queue.api.test.ts (35) 9224ms
✓ tests/lab-03/users-admin.api.test.ts (4) 783ms

Test Files 15 passed (15)
Tests      92 passed (92)
Start at   18:39:17
Duration   28.76s (transform 252ms, setup 0ms, collect 4.96s, tests 18.70s, environment 3ms, prepare 1.77s)

● PS D:\Downloads\toktickit\.capture-main\server> npm run build

> toktickit-server@1.0.0 build
> tsc

○ PS D:\Downloads\toktickit\.capture-main\server> 
```

Original local capture retained under artifacts/lab-03/screenshots/report/26\_terminal\_server\_build.png.

## Evidence 27 — Client build capture

```
PS D:\Downloads\toktickit\capture-main\server> cd D:\Downloads\toktickit\capture-main\client
PS D:\Downloads\toktickit\capture-main\client> npm run build

> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
✓ 37 modules transformed.
dist/index.html          0.41 kB | gzip: 0.27 kB
dist/assets/index-B4cssW3.css 234.03 kB | gzip: 31.83 kB
dist/assets/index-DtMw3Mdc.js 220.95 kB | gzip: 61.26 kB
✓ built in 1.31s
PS D:\Downloads\toktickit\capture-main\client> npx playwright test

Running 13 tests using 1 worker

  ✓ 1 [desktop] › ..\e2e\lab-02\multi-requester-isolation.spec.ts:5:3 › Issue #37 multi-requester isolation › keeps requester A tickets invisible to requester B (2.0s)
  ✓ 2 [desktop] › ..\e2e\lab-02\requester-ticket-flow.spec.ts:8:3 › Issue #37 authenticated requester ticket flow › signs in, creates a ticket, opens detail, and removes an attachment (1.5s)
  ✓ 3 [desktop] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (1.2s)
  ✓ 4 [desktop] › ..\e2e\lab-03\authentication.spec.ts:5:3 › Issue #39 authenticated release journey › completes first-login password setup and shows the Administrator shell (873ms)
  ✓ 5 [desktop] › ..\e2e\lab-03\requester-regression.spec.ts:5:3 › Issue #37 requester communication regression › adds a public comment and records Problem appears resolved without closing the ticket (2.1s)
  ✓ 6 [desktop] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (2.3s)
  ✓ 7 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:5:3 › Issue #38 IT Staff queue and ticket operations › opens the shared queue and staff ticket detail with safe communication controls (1.5s)
  ✓ 8 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:31:3 › Issue #38 IT Staff queue and ticket operations › keeps queue sorting controls usable with mobile ticket cards (613ms)
  ✓ 9 [desktop] › ..\e2e\lab-03\user-administration.spec.ts:5:3 › Issue #39 Administrator user management › lists users and supports create, edit, activation, and initial-password reset (1.5s)
  ✓ 10 [tablet] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (6.4s)
  ✓ 11 [tablet] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (17.2s)
  ✓ 12 [mobile] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (7.6s)
  ✓ 13 [mobile] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (19.1s)

13 passed (1.2m)
PS D:\Downloads\toktickit\capture-main\client>
```

Original local capture retained under artifacts/lab-03/screenshots/report/27\_terminal\_client\_build.png.

## Evidence 28 — Playwright install capture

```
PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\client
● >> npx playwright install --list

Playwright version: 1.62.1
Browsers:
  C:\Users\KEN\AppData\Local\ms-playwright\chromium-1234
  C:\Users\KEN\AppData\Local\ms-playwright\chromium_headless_shell-1234
  C:\Users\KEN\AppData\Local\ms-playwright\ffmpeg-1011
  C:\Users\KEN\AppData\Local\ms-playwright\winldd-1007
References:
  D:\Downloads\toktickit\client\node_modules\playwright-core
  D:\Downloads\toktickit\node_modules\playwright-core
○ PS D:\Downloads\toktickit\client> █
```

Original local capture retained under artifacts/lab-03/screenshots/report/28\_terminal\_playwright\_install.png.

## Evidence 29 — Playwright 13-test capture

```
PS D:\Downloads\toktickit\capture-main\server> cd D:\Downloads\toktickit\capture-main\client
PS D:\Downloads\toktickit\capture-main\client> npm run build

> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
✓ 37 modules transformed.
dist/index.html          0.41 kB | gzip: 0.27 kB
dist/assets/index-B4cssW3.css 234.03 kB | gzip: 31.83 kB
dist/assets/index-DtMw3Mdc.js 220.95 kB | gzip: 61.26 kB
✓ built in 1.31s
PS D:\Downloads\toktickit\capture-main\client> npx playwright test

Running 13 tests using 1 worker

  ✓ 1 [desktop] › ..\e2e\lab-02\multi-requester-isolation.spec.ts:5:3 › Issue #37 multi-requester isolation › keeps requester A tickets invisible to requester B (2.0s)
  ✓ 2 [desktop] › ..\e2e\lab-02\requester-ticket-flow.spec.ts:8:3 › Issue #37 authenticated requester ticket flow › signs in, creates a ticket, opens detail, and removes an attachment (1.5s)
  ✓ 3 [desktop] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (1.2s)
  ✓ 4 [desktop] › ..\e2e\lab-03\authentication.spec.ts:5:3 › Issue #39 authenticated release journey › completes first-login password setup and shows the Administrator shell (873ms)
  ✓ 5 [desktop] › ..\e2e\lab-03\requester-regression.spec.ts:5:3 › Issue #37 requester communication regression › adds a public comment and records Problem appears resolved without closing the ticket (2.1s)
  ✓ 6 [desktop] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (2.3s)
  ✓ 7 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:5:3 › Issue #38 IT Staff queue and ticket operations › opens the shared queue and staff ticket detail with safe communication controls (1.5s)
  ✓ 8 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:31:3 › Issue #38 IT Staff queue and ticket operations › keeps queue sorting controls usable with mobile ticket cards (613ms)
  ✓ 9 [desktop] › ..\e2e\lab-03\user-administration.spec.ts:5:3 › Issue #39 Administrator user management › lists users and supports create, edit, activation, and initial-password reset (1.5s)
  ✓ 10 [tablet] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (6.4s)
  ✓ 11 [tablet] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (17.2s)
  ✓ 12 [mobile] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (7.6s)
  ✓ 13 [mobile] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (19.1s)

13 passed (1.2m)
PS D:\Downloads\toktickit\capture-main\client>
```

Original local capture retained under artifacts/lab-03/screenshots/report/29\_terminal\_playwright\_e2e.png.

## Evidence 30 — Migration and two seed runs

```
PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\capture-main\server
PS D:\Downloads\toktickit\capture-main\server> $line = Get-Content .env.test |
>> Where-Object { $_.StartsWith('DATABASE_URL=') } |
>> Select-Object -First 1
PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> $testUrl = $line.Substring(13).Trim().Trim('"'')
PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> if (-not $testUrl.StartsWith('postgresql://') -and
>> -not $testUrl.StartsWith('postgres://')) {
>> throw "DATABASE_URL is not a PostgreSQL URL"
>> }
PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> if (-not $testUrl.Contains('_test')) {
>> throw "DATABASE_URL is not a test database"
>> }
PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> $env:DATABASE_URL = $testUrl
PS D:\Downloads\toktickit\capture-main\server> Write-Host "Test database loaded; URL hidden." -ForegroundColor Green
Test database loaded; URL hidden.
PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> npx prisma migrate deploy
Prisma schema loaded from prisma/schema.prisma
Datasource "db": PostgreSQL database "toktickit_test", schema "public" at "localhost:5432"

5 migrations found in prisma/migrations

No pending migrations to apply.
PS D:\Downloads\toktickit\capture-main\server> npm run prisma:seed
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeded 4 categories.
Seeded 7 related systems.
Seeded 5 development requesters (4 active, 1 inactive).
Initialized ticket counter.
PS D:\Downloads\toktickit\capture-main\server> npm run prisma:seed
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeded 4 categories.
Seeded 7 related systems.
Seeded 5 development requesters (4 active, 1 inactive).
Initialized ticket counter.
```

Original local capture retained under artifacts/lab-03/screenshots/report/30\_terminal\_migration\_seed.png.

## Submission close-out

The primary deliverable is output/pdf/LAB\_03\_FINAL\_SUBMISSION\_REPORT.pdf. It uses the Lab 2 title/metadata layout, Answer Part headings, gray bordered tables, yellow status callouts, readable spacing and numbered footer throughout. The final report/PDF change is merged into main by PR #49 as b45a853.

No credentials, internal storage keys, filesystem paths or unsupported features are presented as application evidence. The only intentionally simulated UI state is the queue retry capture, explicitly labeled as a Playwright-intercepted 503.